

M.Sc. Applied Mathematics
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Bayesian Clustering

by

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Summary

KPMG Netherlands has provided solution in Audit & Assurance, Advisory, Tax & Legal for over 100 years [9]. The advisory team offers expert guidance on managing risk and improving business performance to support future growth. As part of this mission, Advanced Analytics & Big Data team deals with projects that include leveraging data and AI to design and implement scalable solutions. The team invests in *research and innovation*, allowing them to remain current and deliver impactful solutions across diverse sectors.

Clustering is a method of *unsupervised machine learning* that allows to uncover inherent patterns in complex, high-dimensional datasets, insights that are critical when advising clients across industries such as finance, healthcare, government, retail and more. Common applications include image processing, customer segmentation, and anomaly detection.

The integration of *Bayesian* methods in clustering provides businesses with a framework to handle uncertainty by incorporating prior knowledge into the model. As a result, they allow organizations such as KPMG to make more informed predictions and decisions as such methods produce qualitative results, for example, distribution of possible outcomes, allowing decision-makers to quantify the uncertainty using suitable measures.

Bayesian Clustering is a research master's project conducted in the context of an internship at KPMG Netherlands intended for real-world use. Thus, challenges such as *Performance, Scalability, Interpretability* and *Robustness* must be effectively addressed. In its first component, the project involves the formulation of the theory towards the *Bayesian Clustering for general k-groups*, chapters 1 & 2. The new class of developed algorithms, along with a ready-to-use Python package [bayesiancl] and their performance evaluation can be found in chapters 3 & 4.

Title: Bayesian Clustering

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1 Introduction

In *Machine Learning*, the grouping of objects or data points into "categories" is addressed by two major fields; in supervised learning where labels are provided, *Classification* approaches are used to tackle the already mentioned task, while, *Clustering* is a grouping technique of unsupervised learning where the labels are little or not available in the given data. Consequently, Clustering is more difficult and challenging than Classification. The objective of such analysis; *Clustering analysis* is to develop an automatic algorithm that will discover the natural groupings in the unlabeled, unsupervised data [8].

Moreover, a *Cluster* is vaguely defined as the set of objects with a higher degree of similarity between them than with the objects outside the set [11]. In order to argue the similarities and dissimilarities between data points, suitable measures should be used which consequently give rise to different *clustering approaches*.

1.1 Clustering Approaches

The underlying clustering structure of the various available clustering algorithms serves as a key factor in differentiating between them. Their structure comprises of the choice of the objective function, the probabilistic generative models and the heuristics. Three central and widely used families of clustering approaches are, namely; *Partition-based Clustering*, *Model-Based Clustering*, and *Density-based Clustering* [11].

Partition-based clustering methods, such as K-Means, are known for their efficiency and scalability. They are easy to interpret due to *exclusive cluster membership*, meaning each data point must belong to exactly one cluster. On the other hand, they need to predefine the number of clusters, they are sensitive to initialization of the centroids in the loss function, Table 1.1. In addition, they struggle with non-convex clusters, noise, and outliers.

Density-based clustering defines clusters as dense regions in the data space, separated by regions of lower point density. Exemplified by DBSCAN, it excels in identifying clusters of arbitrary shapes and is robust to noise and outliers. It doesn't require specifying the number of clusters but parameters defining the radius of neighborhood around a point and minimum neighbors within that radius, which are usually domain specific, require careful parameter tuning. As a result DBSCAN can be computationally intensive. Furthermore, it is sensitive to the curse of dimensionality due to its reliance on Euclidean distance measures.

Model-based clustering, such as Gaussian Mixture Models, offers flexibility by modeling data with probabilistic distributions and handling overlapping clusters effectively. It shares the weaknesses of the previously mentioned methods in terms of converging to local optima based on seed assignment and the requirement of specifying the number of clusters a priori. Additionally, the optimization criteria can theoretically take infinite iterations to reach convergence. While it provides valuable interpretability through soft clustering: each point gets a probability distribution over clusters, it can be computationally demanding and less scalable, especially for large datasets.

Partition-Based The method assumes initial centroids. The objective is to minimize the mean squared error loss function with respect to the centroids, which are usually, either the mean or the median of the points. Each centroid represents each cluster [13]. $\sum_{j=1}^k \sum_{i=1}^n \ x_i^{(j)} - c_j\ ^2$ where $\ x_i^{(j)} - c_j\ ^2$ is chosen as the distance measure of point x_i from centroid c_j.	Model-Based A GMM assumes that the data points $x_1, x_2, \dots, x_n$ are generated from a mixture of K Gaussian distributions each with its own mean vector μ_k and covariance matrix Σ_k. $p(x_i \Theta) = \sum_{k=1}^K \pi_k \cdot \mathcal{N}(x_i \mu_k, \Sigma_k)$ The objective is to maximize, $L(\theta) = \sum_{i=1}^N \log \left(\sum_{k=1}^K \pi_k \mathcal{N}(x_i \mu_k, \Sigma_k) \right)$
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Table 1.1: Partition-Based and Model-Based Clustering

Ultimately, selecting the most appropriate method depends on the specific characteristics of the data and the goals of the analysis, taking into account trade-offs be-

tween efficiency, robustness, scalability, and complexity — including challenges such as predefining the number of clusters or other parameters, sensitivity to initialization as well as convergence.

This study employs a model-based clustering approach, which will be evaluated in comparison with partition-based (KMeans) and density-based (DBSCAN) methods in Section ?? . Before introducing the method in detail, we begin by establishing some preliminary notation and concepts.

1.2 Notation

Following the notation as given in [3] and [4]. For simplicity, we often skip the dependence of every measure, i.e. P_θ , on σ and n . Denote by $|S|$ the cardinality of the set S , the difference of sets $S \setminus S_0 = \{s \in S : s \notin S_0\}$, $[n] = \{1, \dots, n\}$ for $n \in \mathbb{N}$, $[n]_0 = \{0\} \cup [n]$, $\mathbb{N}_0 = \{0\} \cup \mathbb{N}$. For a $S \subseteq [n]$ and a vector $v = (v_i, i \in [n]) \in \mathbb{R}^n$, define $S^c = [n] \setminus S$ and the sub-vector $v_S = (v_i, i \in S) \in \mathbb{R}^{|S|}$. Throughout $Z \sim \mathcal{N}(0, 1)$ will denote a generic standard normal random variable, with distribution function $\Phi(z) = P(Z \leq z)$ and density $\phi(z) = \Phi'(z)$. Let $\phi(x, \mu, \sigma^2)$ be the density of $\mu + \sigma Z \sim \mathcal{N}(\mu, \sigma^2)$ at point x .

1.2.1 Partition Space

A single clustering can be viewed as partitioning the indices of the points $[n] = \{1, \dots, n\}$ into k distinct sets. Similarly to the notation in [14], a single partition can be represented either by the set $I = (I_1, \dots, I_k)$, where I_j contains the indices of data points in the j^{th} cluster and k is the number of clusters in the sample of size n . Alternatively, the partition can be represented by, namely, the labels, $c = (c_1, \dots, c_n)$, where $c_n = j$ if the n^{th} data point is in the j^{th} cluster.

The partition space $\mathcal{I}_k$ refers to the family of all possible partitions of points n into $l \in [k]$ disjoint groups, (1.1). By definition, it holds that $\mathcal{I}_k = \cup_{l \in [k]} \mathcal{I}_l$, for $k \in [n]$ and further the cardinality of family $\mathcal{I}_k$ is $|\mathcal{I}_k| = k^n$.

$$\mathcal{I}_k = \{I = (I_1, \dots, I_k) : \cup_{i=1}^k I_i = [n] \text{ and } I_i \cap I_j = \emptyset \text{ for all } i, j \in [n] : i \neq j\}. \quad (1.1)$$

Lastly, abusing the notation is introduced for $\mathbf{m} = |I|$ of partition $I = (I_1, \dots, I_k) \in \mathcal{I}_k$, to be a cardinality-vector of the respective cardinalities of groups $|I_1|, \dots, |I_n|$.

1.3 Signal plus noise model

Suppose that we observe a vector $X = (X_1, \dots, X_n) \in \mathbb{R}^n$ such that

$$X_i = \theta_i + \epsilon_i, \quad i = 1, \dots, n \quad (1.2)$$

for independent random errors ϵ_i and an unknown vector of interest $\theta = (\theta_1, \theta_2, \dots, \theta_n) \in \mathbb{R}^n$. It is further assumed that the errors; $\mathbb{E}(\epsilon_i) = 0$ and $\mathbb{V}(\epsilon_i) = \sigma_i$, $i = 1, \dots, n$ and thus, θ is a vector of means [5].

1.4 Sparsity

An intuitive characterization of sparsity is provided by the concept of *nearly black vectors*. In this setting, sparsity can be defined in terms of the ℓ_0 -ball and has just one known sparsity level which is, by default, zero. This is given by,

$$\ell_0(p) = \{\theta \in \mathbb{R}^n : \#\{i \in \{1, \dots, n\} : \theta_i \neq 0\} \leq p\}$$

where p/n is a fixed proportion of non-zero entries [1].

Taking into account the model (1.2) along with the traditional sparsity structure, it follows that $\theta \in \ell_0(p)$, indicating that θ has only a small proportion of nonzero coefficients and can thus be regarded as a sparse vector. The sparsity class can be expressed by the linear subspaces,

$$\mathbb{L}_o = \{\theta \in \mathbb{R}^n : \theta_i = \theta_j = 0, \forall i, j \in I_o^c\}, \quad I = (I_o, I_o^c) \in \mathcal{I}_2.$$

Allowing the value of the sparse cluster to be unknown [6] the vector of interest becomes $\theta = (\theta_I, \theta_{I^c})$ and similarly,

$$\mathbb{L}_I = \{\theta \in \mathbb{R}^n : \theta_i = \theta_j, \forall i, j \in I_o^c\}, \quad I \in \mathcal{I}_2. \quad (1.3)$$

Extending the above traditional one-level sparsity to namely the multi-level sparsity as in [4]. It holds that, $\theta \in l_0(p) \subseteq \mathbb{R}^n$ and the goal is to partition its coordinates into $p+1$ groups: one large group consisting of all zero-valued coordinates $\{i \in [n] : \theta_i = 0\}$ and p distinct groups, each containing one of the significant coordinates $\{i \in [n] : \theta_i \neq 0\}$.

0}. By allowing the sparsity level to be an *unknown fixed constant* rather than strictly zero, the traditional sparsity structure is generalized to sparsity defined by linear subspaces of the form,

$$\mathbb{L}_{\mathcal{I}_k} = \left\{ \theta \in \mathbb{R}^n : \theta_j = \theta_{j'}, \forall j, j' \in I_l, \forall l \in [k] \right\}, \quad (I_1, \dots, I_k) \in \mathcal{I}_k.$$

1.5 Bayesian Inference

The frequentist view defines the probability of an event as the proportion of times that the event occurs in a sequence of possibly hypothetical trials. [2]

In the frequentist approach one will determine the corresponding parameter (i.e. mean) through the marginal likelihood or the method of moments. In the Bayesian framework, instead for picking fixed values, the parameters of interest are being treated as *random variables* which are generated by the so-called *prior* distribution.

The prior distribution expresses one's belief for each $\theta \in \Theta$ which differs from the frequentist approach; probability of an event is not approximated by repetition of the same experiment again and again.

Bayesian inference is inherently conditional on a specific observed sample. This approach can be vague in cases where the belief we hold about the underlying distribution is little or non existent.

Consider the random sample $X_1, \dots, X_n \sim f(x|\theta)$ and our belief on the prior $\theta \sim \pi(\theta)$. Intuitively this means that, θ is more likely to be around certain values and less likely around others. After those random variables have been realized; the observed data, $x_1, x_2, \dots, x_n$ are drawn from distribution $f(x|\theta)$. The probability of observing θ given $x_1, x_2, \dots, x_n$ according to *Baye's Theorem*, namely, the posterior distribution $\pi(\theta|x)$ is defined as,

$$\pi(\theta|x) = \frac{f(x|\theta)\pi(\theta)}{\int_{\Theta} f(x|\theta)\pi(\theta) d\theta} \quad (1.4)$$

where $\int_{\Theta} f(x|\theta)\pi(\theta) d\theta$ is the marginal distribution and $f(x|\theta)$ is the likelihood. Note that $\int_{\Theta} f(x|\theta)\pi(\theta) d\theta$ is independent of θ and acts as a normalizing constant. Consequently,

$$\pi(\theta|x) \propto f(x|\theta)\pi(\theta)$$

In other words, $\pi(\theta|x)$ is the *a posteriori* belief one holds for the nature/distribution of the parameter of interest θ after observing the given data.

1.6 Hierarchy in Bayesian Clustering

The Bayesian model can be described by the two-step algorithm to generate a data point x and a parameter θ as follows,

Step 1: θ generated by $\pi(\theta)$

Step 2: $X_1, \dots, X_n$ generated by $f(x|\theta)$

Here, x is the *observed* data whereas θ is *unobserved* parameter [12]. In case the unobservable θ consists of other variables besides the parameter, another level of hierarchy is introduced and the generative model will consist of more than two steps.

Step 1: α generated by $\phi(\alpha)$

Step 2: θ generated by $\pi_\theta(\alpha)$ (1.5)

Step 3: $X_1, \dots, X_n$ generated by $f(x|\theta)$

In this hierarchy, every unobservable receives a distribution and after observing the data the distributions are updated to their conditional distributions given the data.

Theoretically, one can include as many hierarchy levels as needed. In the clustering context, two hierarchy levels will be assumed as given in (1.5) where the random variables are; the data points x , the high dimensional vector of interest θ and the partition I . The algorithm becomes,

Step 1: I generated by λ_I

Step 2: θ generated by Π (1.6)

Step 3: $X_1, \dots, X_n$ generated by $P(x|\theta)$

where λ_I and Π are the selected priors distributions for the model. The analysis of the structure of the random variables and their distributions is done in the next chapter.

2 Model

Suppose observed data, $X = (X_1, \dots, X_n) \in \mathbb{R}^n$ generated by $X \sim P_\theta$, it must hold that,

$$X_i \sim P_{\theta_i} \quad i = 1, \dots, n.$$

and,

$$X \sim P_\theta = \bigotimes_{i=1}^n P_{\theta_i} \quad (2.1)$$

where $\theta = (\theta_1, \dots, \theta_n) \in \mathbb{R}^n$ is a high-dimensional vector of interest, and $\bigotimes_{i=1}^n P_{\theta,i}$ denotes *the product measure* of the individual measures.

Furthermore, the structure of θ must be such that it induces the notion of the "cluster".

2.1 Clustering structure

It is possible to derive the clustering structure of θ from the multi-level sparsity as an extension of the previously defined simple sparsity (1.4). In this case, the family of all partitions of $[n]$ in k groups, is given by,

$$\mathcal{I}_k = \left\{ I : \bigcup_{i=1}^k I_i = [n], I_j \cap I_{j'} = \emptyset, \forall j, j' \in [k] : j \neq j' \right\}$$

where a partition $I = (I_i, i \in [k])$ of $[n]$ into k groups. Then θ is an element of the following linear subspaces,

$$\mathbb{L}_I = \{ \theta \in \mathbb{R}^n : \theta_i = \theta_j, \forall i, j \in I_i, i \in [k] \}, \quad I \in \mathcal{I}_k \quad (2.2)$$

Using the simple one-level sparsity structure together with the linear subspaces approach, the task is transformed into grouping the coordinates of θ into k distinct

groups. Hence, multi-level sparsity can be viewed as a *clustering structure* for θ .

2.2 Bayesian Clustering for known partition

The focus is on the main three components of the Bayesian Analysis, as mentioned in section (1.5); *the prior, the marginal likelihood and the posterior* in order to formulate the bayesian clustering model for known partition I .

Lemma 2.1. *Consider observations $X = (x_1, \dots, x_n) \in \mathbb{R}^n$, the model (??) and the structure of $\theta \in \mathbb{L}_I$ as given in (2.2) for a known partition $I = (I_1, \dots, I_k) \in \mathcal{I}_k$. Assuming the prior $\theta \sim \Pi_I$, it follows that the prior can be expressed as,*

$$\Pi_I = \bigotimes_{i \in [n]} \Pi_{I,i} = \bigotimes_{l \in [k]} \bigotimes_{i \in I_l} \Pi_{I_l,i}.$$

In the same fashion, for the marginal likelihood, it holds that,

$$P_{X,I} = \bigotimes_{i \in [n]} P_{X,I,i}(x_i) = \bigotimes_{l \in [k]} \bigotimes_{i \in I_l} P_{X,I_l,i}(x_i)$$

where,

$$P_{X,I,i}(x_i) = \int P_{\theta_i}(x_i) d\Pi_{I,i}(\theta_i) = \int P_{\theta_i}(x_i) \Pi_{I,i} d\theta_i \quad (2.3)$$

Lastly, for the posterior,

$$\Pi_I(\theta | X) = \bigotimes_{i \in [n]} \Pi_{I,i}(\theta_i | X_i) = \bigotimes_{l \in [k]} \bigotimes_{i \in I_l} \Pi_{I_l,i}(\theta_i | X_i)$$

where $\forall l \in [k]$ and $\forall i \in I_l$, it holds that,

$$\Pi_{I,i}(\theta_i | X_i) = \frac{\Pi_{I,i}(\theta_i) P_{\theta_i}(X_i)}{P_{X,I,i}(X_i)}, \text{ and } \Pi_{I_l,i}(\theta_i | X_i) = \frac{\Pi_{I_l,i}(\theta_i) P_{\theta_i}(X_i)}{P_{X,I_l,i}(X_i)}.$$

Proof. The posterior expression follows directly from the Bayesian Theorem (1.4). $\square$

2.3 Bayesian Clustering for unknown partition

Naturally, the objective of clustering is to identify an optimal (based on some criteria) grouping of the observations; therefore, the partition I should be considered unknown a priori. To account for the unknown partition I , we assign a prior to the family of all partitions, thus one more hierarchical level is introduced to the model.

Remark. Let λ be a prior on the partition space $\mathcal{I}_k$, $I \sim \lambda_I$. It follows that a mixture prior on θ is obtained and it takes the form, $\Pi = \sum_{I \in \mathcal{I}_k} \lambda_I \Pi_I$,

$$\theta \sim \sum_{I \in \mathcal{I}_k} \lambda_I \Pi_I.$$

and the Bayesian Clustering model is expressed by,

$$\begin{aligned} X &\sim P_\theta \\ \theta &\sim \sum_{I \in \mathcal{I}_k} \lambda_I \Pi_I \\ I &\sim \lambda_I \end{aligned} \tag{2.4}$$

Lemma 2.2. *Let observations $X = (X_1, \dots, X_n) \in \mathbb{R}^n$ and the Bayesian Clustering model (2.4). The triplet of bayesian components are rewritten in the following way, The marginal likelihood,*

$$P_X = \sum_{I \in \mathcal{I}_k} \lambda_I P_{X,I} = \sum_{I \in \mathcal{I}_k} \lambda_I \bigotimes_{l \in [k]} \bigotimes_{i \in I_l} P_{X,I_l,i}(x_i),$$

the I -posterior, $\Pi(I|X)$,

$$\Pi(I|X) = \frac{\lambda_I P_{X,I}}{P_X} = \frac{\lambda_I}{P_X} \bigotimes_{l \in [k]} \bigotimes_{i \in I_l} P_{X,I_l,i}(x_i).$$

and, the θ -posterior $\Pi(\theta|X)$,

$$\Pi(\theta|X) = \sum_{I \in \mathcal{I}_k} \Pi_I(\theta|X) \Pi(I|X) = \sum_{I \in \mathcal{I}_k} \frac{\lambda_I}{P_X} \bigotimes_{l \in [k]} \bigotimes_{i \in I_l} \Pi_{I_l,i}(\theta_i) \cdot P_{\theta_i}(x_i).$$

The prior $\Pi = \sum_{I \in \mathcal{I}_k} \lambda_I \Pi_I$ remains the same.

The proof can be found in Appendix A.1.

The focus must be on the posterior $\Pi(I|X)$ since the clustering goal is to estimate partition I . The subsequent sections extend the analysis from the case of two-group clustering to the general case k -groups clustering, introducing the *product-form prior* and *cardinality-based prior* based on the generation of the mixture weights λ_I 's.

2.4 Clustering in 2-groups

Similar to the sparsity setting, as given by (1.3), where two clusters are formed; *on* and *off* cluster, working with a single cluster is sufficient to determine the entire partition. Without loss of generality, I_o is chosen to determine partition I , $I = (I_o, I_o^c)$. Thus, we continue by assigning weights to individual points of cluster I_o .

2.4.1 Prior in product-form

Definition 2.3. Let $X = (X_1, \dots, X_n) \in \mathbb{R}^n$ and consider the model (2.1). Set the individual weights for cluster I_o ,

$$\lambda_i, \quad \forall i \in [n].$$

Then, the *product-form* prior $I \sim \lambda_I^{prod}$ on the partition space $\mathcal{I}_2$ is defined as,

$$\lambda_I^{prod} := C_\lambda \prod_{i \in I_o} \lambda_i, \quad C_\lambda = \prod_{j \in [n]} (1 + \lambda_j)^{-1},$$

where C_λ acts as a normalizing constant.

The derivation of the normalizing constant is provided in the Appendix A.1. In this approach, the prior $I \sim \lambda_I$ is of the *product-form* of individual weights per group-point combination. Considering the representation of the product-form prior to be, $\lambda_I = [\lambda_1, \dots, \lambda_n] \in \mathbb{R}^{1 \times n}$ as the prior only depends on I_o .

Theorem 2.4. Let a sample $X = (X_1, \dots, X_n) \in \mathbb{R}^n$, the clustering setting for $\theta \in \mathbb{L}_{\mathcal{I}_2}$ and the model (2.4). Suppose product-form prior on $\mathcal{I}_2$, it follows that the prior on θ , the

marginal likelihood and the posteriors for I and θ , respectively, can be reformulated as,

$$\begin{aligned}\Pi &= \bigotimes_{j \in [n]} \left[\omega_j \Pi_{1,j} + (1 - \omega_j) \Pi_{2,j} \right], \quad P_X = \bigotimes_{j \in [n]} \left[\omega_j P_{X,1,j}(x_j) + (1 - \omega_j) P_{X,2,j}(x_j) \right], \\ \Pi(I|X) &= \frac{C_\lambda}{P_X} \bigotimes_{j_1 \in I_o} \omega_{j_1} P_{X,1,j_1}(x_{j_1}) \times \bigotimes_{j_2 \in I_o^c} P_{X,2,j_2}(x_{j_2}), \\ \Pi(\theta|X) &= \frac{1}{P_X} \bigotimes_{j \in [n]} \left[\omega_j \Pi_{1,j}(\theta_j) P_{\theta_j}(x_j) + (1 - \omega_j) \Pi_{2,j}(\theta_j) P_{\theta_j}(x_j) \right],\end{aligned}$$

where, $\omega_j = \frac{\lambda_j}{1 + \lambda_j}$, $j \in [n]$ are the posterior weights.

The proof is provided in the Appendix A.1.

Connetion between methods

Remark. It follows from the derivation of the bayesian formulas, Theorem 2.4 that, $\omega_j = \mathbb{P}(j \in I_o)$.

Remark. Suppose $\omega_i = \mathbb{P}(i \in I_o) = p$, $\forall i \in [n]$ then the $\mathbb{P}(I_o) = p^{|I_o|}(1 - p)^{(n - |I_o|)}$ where $m := |I_o|$ is the cardinality of group I_o .

2.4.2 Cardinality-based prior

In this case, we deviate from the product-form assumption and introduce a prior on the cardinality of group I_o .

Definition 2.5. Let a sample $X = (X_1, \dots, X_n) \in \mathbb{R}^n$ in the clustering setting for $\theta \in \mathbb{L}_{\mathcal{I}_2}$ and the model (2.4). The *cardinality-based* prior λ^{card} s.t. $I \sim \lambda_I^{card}$ is defined as follows,

$$\lambda_I^{card} := g(m) \binom{n}{m}^{-1}, \quad m \in \{0, \dots, n\}$$

where m is the cardinality of group $I_o \in I$, $m = |I_o|$ and $g(\cdot)$ nonnegative function on $[n]_0$ s.t. $\sum_{m \in [n]} g(m) = 1$.

The construction of the cardinality-based prior is such that, $\sum \lambda_I^{prod} = 1$, for the exact derivation refer to the Appendix A.1.

Theorem 2.6. Let a sample $X = (X_1, \dots, X_n) \in \mathbb{R}^n$ in the clustering setting for $\theta \in \mathbb{L}_{\mathcal{I}_2}$ and the cardinality-based prior as given by definition (2.5), it follows that the components for bayesian analysis; θ -prior, the marginal likelihood and the posteriors for I and θ , respectively, are expressed as,

$$\Pi = \sum_{m=0}^n g(m) A_m(\Pi_1, \Pi_2), \quad P_X = \sum_{m=0}^n g(m) A_m(P_{X,1}(x), P_{X,2}(x)),$$

$$\text{and } \Pi(\theta|X) = \frac{\sum_{m=0}^n g(m) A_m(\rho_1(x), \rho_2(x))}{P_X} \quad \text{with} \quad \rho_l = \Pi_{l,i}(\theta_i) P_{\theta_i}(x), \quad l = 1, 2.$$

For simplicity Π_1, Π_2 denotes $\Pi_{I_0}, \Pi_{I_0^c}$ respectively and similarly for $P_{X,1}$ and $P_{X,2}$. The term $A_m(\cdot)$ represents the m^{th} coefficient of the polynomial as given in (the following) Lemma 2.7.

The expression of $\Pi(I|X)$ retains its original form (2.2).

Proof. Following the definition (2.5) for the cardinality-based prior, the prior on θ (Π), the marginal likelihood P_X and the posteriors $\Pi(\theta|X)$, can be rewritten as,

$$\Pi = \sum_{I \in \mathcal{I}} \lambda_I \Pi_I = \sum_{m=0}^n \sum_{I: |I_0|=m} g(m) \left(\bigotimes_{i \in I_0} \Pi_{1,i} \times \bigotimes_{j \in I_0^c} \Pi_{2,j} \right), \quad (2.5)$$

$$P_X = \sum_{I \in \mathcal{I}} \lambda_I P_{X,I} = \sum_{m=0}^n \sum_{I: |I_0|=m} \left(g(m) \bigotimes_{i \in I_0} P_{X,1,i}(x_i) \times \bigotimes_{j \in I_0^c} P_{X,2,j}(x_j) \right), \quad (2.6)$$

$$\Pi(\theta|X) = \frac{\sum_{m=0}^n \sum_{I: |I|=m} g(m) \bigotimes_{i \in I_0} \Pi_{1,i}(\theta_i) P_{\theta_i}(x_i) \times \bigotimes_{j \in I_0^c} \Pi_{1,j}(\theta_j) P_{\theta_j}(x_j)}{P_X}. \quad (2.7)$$

As a preliminary step toward deriving the associated expressions, we make use of the polynomial trick.

Lemma 2.7 (Polynomial trick [7]). Let $X = (X_1, \dots, X_n) \in \mathbb{R}^n$. First, define the function space as $\mathcal{F} = \{f \in C^2(\mathbb{R})\}$. Then define the functional $\Phi: \mathcal{F} \times \mathcal{F} \rightarrow \mathbb{R}$ by

$$\Phi[\psi, \phi] = \prod_{i=1}^n (\psi(x_i) + \phi(x_i) Z)$$

where $Z \in \mathbb{R}$ (or $\mathbb{C}$) is the polynomial variable. It holds that, the m^{th} coefficient of the polynomial $Z \rightarrow \Phi[\psi, \phi](Z)$, denoted by $A_m(\psi(x), \phi(x))$ is equal to the quantity,

$$Q_n = \sum_{|S|=m} \prod_{i \in S} \psi(x_i) \prod_{i \notin S} \phi(x_i)$$

where $S \in \mathcal{I}_2$.

The proof can be found in the Appendix A.1.

Suppose $\psi = \Pi_1$ and $\phi = \Pi_2$ (indep. of x). By applying the polynomial trick on the expression (2.5) for Π , it follows that,

$$\Pi = \sum_{m=0}^n g(m) \sum_{I: |I_o|=m} \left(\bigotimes_{i \in I_o} \Pi_{1,i} \times \bigotimes_{j \in I_o^c} \Pi_{2,j} \right) = \sum_{m=0}^n g(m) \cdot Q_n = \sum_{m=0}^n g(m) A_m(\Pi_1, \Pi_2).$$

Similarly, suppose the pairs $(\psi = P_{X,1}, \phi = P_{X,2})$ and $(\psi = \rho_1, \phi = \rho_2)$ where $\rho_l = \Pi_{l,i}(\theta_i) P_{\theta_i}$ for $l = 1, 2$, for P_X and the posterior, respectively. By applying the polynomial trick for the expressions (2.6), (2.7), we arrive at the required result. $\square$

In the following section, we aim to extend from 2-groups to general k-groups.

2.5 Clustering into k-groups

2.5.1 Prior in product-form

Similarly to the multi-level sparsity for unknown cluster values [??], we extend into general clustering into k-groups. Without loss of generality and since $k-1$ groups are sufficient to determine the entire partition,

$$I = (I_1, I_2, \dots, I_{k-1}, I_k), \quad I_k = [n] \setminus \bigcup_{l=1}^{k-1} I_l$$

we continue by assigning weights to $k-1$ groups.

Definition 2.8. Let sample $X = (X_1, \dots, X_n) \in \mathbb{R}^n$, partition $I \in \mathcal{I}_k$ and the individual prior weights for the $k-1$ groups,

$$\lambda_{l,j}, \quad \forall j \in [n], l \in [k-1].$$

Then the *product-form* prior $I \sim \lambda_I^{prod}$, on $\mathcal{I}_k$ is defined by,

$$\lambda_I^{prod} = C_\lambda \prod_{i \in I_1} \lambda_{1,i} \cdots \prod_{i \in I_{k-1}} \lambda_{k-1,i} \text{ or, equivalently, } \lambda_I^{prod} = C_\lambda \prod_{l \in [k-1]} \prod_{i \in I_l} \lambda_{l,i}$$

where the normalizing constant $C_\lambda := \prod_{i=1}^n (\lambda_{1,i} + \cdots + \lambda_{k-1,i} + 1)^{-1}$.

The construction of C_λ can be found in the Appendix A.1. The expressions used in the bayesian analysis are adapted to k-groups in the following way.

Theorem 2.9. *Let a sample $X = (X_1, \dots, X_n) \in \mathbb{R}^n$, the clustering setting for $\theta \in \mathbb{L}_{\mathcal{I}_k}$ and the model (2.4). Suppose the prior is of product-form. It follows that, the prior, the marginal likelihood and the posteriors for I and θ , respectively, can be reformulated as,*

$$\begin{aligned} P_X &= \bigotimes_{i \in [n]} \left[\sum_{l \in [k-1]} \omega_{l,i} P_{X,I_l,i}(x_i) + \left(1 - \sum_{l \in [k-1]} \omega_{l,i}\right) P_{X,I_k,i}(x_i) \right], \\ \Pi &= \bigotimes_{i \in [n]} \left[\sum_{l \in [k-1]} \omega_{l,i} \Pi_{I_l,i}(x_i) + \left(1 - \sum_{l \in [k-1]} \omega_{l,i}\right) \Pi_{I_k,i}(x_i) \right], \\ \Pi(I|X) &= \frac{C_\lambda}{P_X} \bigotimes_{l \in [k-1]} \bigotimes_{i \in I_l} \lambda_{l,i} P_{X,I_l,i}(x_i) \times \bigotimes_{j \in I_k} P_{X,I_k,j}(x_j), \end{aligned}$$

and,

$$\Pi(\theta|X) = \frac{1}{P_X} \bigotimes_{i \in [n]} \left[\sum_{l \in [k-1]} \omega_{l,i} \Pi_{I_l,i}(\theta_i) \cdot P_{\theta_i}(X_i) + \left(1 - \sum_{l \in [k-1]} \omega_{l,i}\right) \Pi_{I_k,i}(\theta_i) \cdot P_{\theta_i}(X_i) \right].$$

The posterior weights $\omega_{l,i}$ are given by,

$$\omega_{l,j} = \frac{\lambda_{l,j}}{\sum_{s=1}^{k-1} \lambda_{s,j} + 1}, \quad l \in [k-1], j \in [n].$$

The proof is provided in the Appendix A.1.

2.5.2 Cardinality-based prior

In the generalized case, the cardinality prior will depend on the cardinalities of $k-1$ groups.

Definition 2.10. Let a sample $X = (X_1, \dots, X_n) \in \mathbb{R}^n$ in the clustering setting for $\theta \in \mathbb{L}_{\mathcal{I}_2}$, the model (2.4) and a random partition $I \in \mathcal{I}_k$. The *cardinality-based* prior for k groups λ^{card} s.t. $I \sim \lambda_I^{card}$ is defined by,

$$\lambda_I = c_{\mathbf{m}} g(\mathbf{m}), \quad \text{with } c_{\mathbf{m}} = \binom{n}{m_1, m_2, \dots, m_k}^{-1}.$$

where $\mathbf{m} = (m_1, \dots, m_k)$ are the cardinalities of groups $I_1, \dots, I_k$ and $g(\cdot) : \mathcal{M} \rightarrow [0, 1]$, is a nonnegative function s.t. $\sum_{\mathbf{m} \in \mathcal{M}} g(\mathbf{m}) = 1$,

$$\mathcal{M} = \{\mathbf{m} = (m_1, \dots, m_k) \in \mathbb{N}_0^k : m_1 = |I_1|, \dots, m_k = |I_k|\}.$$

The construction of $I \sim \lambda_I^{card}$ can be found in the Appendix A.1.

Theorem 2.11. Let a sample $X = (X_1, \dots, X_n) \in \mathbb{R}^n$ in the clustering setting for $\theta \in \mathbb{L}_{\mathcal{I}_2}$ and the cardinality-based prior $I \sim \lambda_I^{card}$. It follows that the components for bayesian analysis; θ -prior, the marginal likelihood and the posteriors for I and θ , respectively, are expressed by,

$$\begin{aligned} \Pi &= \sum_{\mathbf{m} \in \mathcal{M}} g(\mathbf{m}) A_{\mathbf{m}}(\Pi_1, \Pi_2, \dots, \Pi_k), \quad P_X = \sum_{\mathbf{m} \in \mathcal{M}} g(\mathbf{m}) A_{\mathbf{m}}(P_{X,1}(x), P_{X,2}(x), \dots, P_{X,k}(x)) \\ \Pi(\theta|X) &= \frac{\sum_{\mathbf{m} \in \mathcal{M}} \sum_{I:|I|=\mathbf{m}} g(\mathbf{m}) A_{\mathbf{m}}(\rho_1(x), \rho_2(x), \dots, \rho_k(x))}{P_X} \quad \text{with } \rho_l = \Pi_{I_l}(\theta) P_{\theta}(x), \quad l \in [k]. \end{aligned}$$

The term $A_{\mathbf{m}}(\cdot)$ represents the m^{th} coefficient of the polynomial as given in (the following) Lemma 2.12.

The expression of $\Pi(I|X)$ retains its original form (2.2).

Proof. The marginal, the prior and the posterior, are expressed respectively,

$$P_X = \sum_{I \in \mathcal{I}_k} \lambda_I \bigotimes_{l \in [k]} \bigotimes_{i \in I_l} P_{X, I_l, i}(x_i) = \sum_{\mathbf{m} \in \mathcal{M}} \sum_{I: |I|=\mathbf{m}} \left(g(\mathbf{m}) \bigotimes_{l \in [k]} \bigotimes_{i \in I_l} P_{X, I_l, i}(x_i) \right), \quad (2.8)$$

$$\Pi = \sum_{I \in \mathcal{I}} \lambda_I \Pi_I = \sum_{\mathbf{m} \in \mathcal{M}} \sum_{I: |I|=\mathbf{m}} \left(g(\mathbf{m}) \bigotimes_{l \in [k]} \bigotimes_{i \in I_l} \Pi_{I_l, i} \right), \quad (2.9)$$

$$\Pi(\theta|X) = \sum_{I \in \mathcal{I}_k} \lambda_I \frac{\bigotimes_{l \in [k]} \bigotimes_{i \in I_l} \Pi_{I_l, i}(\theta_i) \cdot P_{\theta_i}(x_i)}{P_X}. \quad (2.10)$$

The functional for k-groups is going to be generalized in the following multivariate polynomial.

Lemma 2.12 (Multivariate polynomial trick). *Let $X = (X_1, \dots, X_n) \in \mathbb{R}^n$. First, define the function space as $\mathcal{F} = \{f \in C^2(\mathbb{R})\}$. Then define the functional $\Phi: \mathcal{F}^k \rightarrow \mathbb{R}$ by,*

$$\Phi[\psi_1, \dots, \psi_k] = \prod_{i=1}^n (\psi_1(x_i)Z_1 + \psi_2(x_i)Z_2 + \dots + \psi_k(x_i)Z_k),$$

where, $Z_1, \dots, Z_k \in \mathbb{R}$ (or $\mathbb{C}$) are the polynomial variables. By setting $Z_1 \equiv 1$, the coefficient of order $\mathbf{m} = (m_1, m_2, \dots, m_k)$ of the multivariate polynomial $Z = (Z_1, \dots, Z_k) \rightarrow \Phi[\psi_1, \dots, \psi_k](Z)$, denoted by $A_{\mathbf{m}}(\psi_1(x), \dots, \psi_k(x))$ is equal to the quantity,

$$R_n = \sum_{|S|=\mathbf{m}} \prod_{i \in S_1} \psi_1(x_i) \prod_{i \in S_2} \psi_2(x_i) \cdots \prod_{i \in S_k} \psi_k(x_i),$$

where $S = (S_1, \dots, S_k) \in \mathcal{I}_k$.

The proof can be found in the Appendix A.1.

For P_X as given by expression (2.8), by setting $\psi_1 = P_{X, I_1}, \dots, \psi_k = P_{X, I_k}$ and using the Multivariate Polynomial trick, we arrive at the required expression,

$$P_X = \sum_{\mathbf{m} \in \mathcal{M}} g(\mathbf{m}) A_{\mathbf{m}}(P_{X, I_1}(x), P_{X, I_2}(x), \dots, P_{X, I_k}(x)).$$

In the same fashion, for Π (independent of x), by setting $\psi_1 = \Pi_{I_1}, \dots, \psi_k = \Pi_{I_k}$ and lastly, for $\Pi(\theta|X)$, $\psi_l = \rho_l$ where $\rho_l = \Pi_{I_l}(\theta)P_{\theta_i}, \forall l \in [k]$, by using the multivariate polynomial trick for expressions (2.9), (2.10), respectively, we arrive at the required ex-

pressions. □

2.6 Prediction

The posterior $\Pi(I|X)$ is used for inference on the partition of θ into k groups of coordinates. To quantify the posterior the *Maximum A-Posteriori* (MAP) estimator is proposed,

$$\hat{I}_{MAP} = \arg \max_{I \in \mathcal{I}_k} \Pi(I|X) = \arg \max_{I \in \mathcal{I}_k} \lambda_I P_{X,I}.$$

An approach to derive such an estimator is the **brute-force** setting of parsing the partition space, as demonstrated in Algorithm 2 for the cardinality based prior and Algorithm 1 for the product-form prior. However, a major drawback of this methodology is that the size of the partition space grows exponentially (k^n) with the number of observations n , making such algorithms computationally infeasible for moderate to large datasets. Thus, an alternative approach should be introduced.

1. Product-form prior: Partition Space

Input: Data $x_1, \dots, x_n \sim P_\theta$, matrix $\Lambda = (\lambda_{li}) \in \mathbb{R}^{(k-1) \times N}$, prior means $\mu_1, \dots, \mu_k$ and prior $\theta \sim \Pi_\theta$

Output: Partition Estimator $\hat{I}$ // MAP-Estimator

- 1: Generate the partition space $\mathcal{I}_k$
 - 2: **for** each $l \in [k]$ **do**
 - 3: **for** each $i \in [n]$ **do**
 - 4: Compute $a_l \leftarrow (P_{X,l,i}(x_i))_{l,i}$
 - 5: **end for**
 - 6: **end for**
 - 7: Compute normalizing constants, C_λ, P_X
 - 8: Compute posterior mixture weights $\omega_{li} = \frac{\lambda_{li}}{\sum_{s=1}^{k-1} \lambda_{si} + 1}$, and $\omega_{ki} = 1 - \sum_{s=1}^{k-1} \omega_{si}$
 - 9: **for** each $I \in \mathcal{I}_k$ **do**
 - 10: Compute $\lambda_I = C_\lambda \cdot \prod_{l \in [k]} \prod_{i \in I_l} \lambda_{li}$
 - 11: Compute marginal likelihood $P_{X,I}(x) = \prod_{l \in [k]} \prod_{i \in I_l} P_{X,l,i}(x_i)$
 - 12: Store product $\Pi(I|X) = \frac{\lambda_I \cdot P_{X,I}(x)}{P_X}$
 - 13: **end for**
 - 14: **return** Partition $\hat{I} = \arg \max_I \Pi(I|x)$
-

Remark. Due to floating-point precision limits, repeated multiplication of the prior weights λ can cause the values to vanish. The constants P_X and C_λ counteract this by stabilizing the computation.

2. Cardinality-Based Prior: Partition Space

Input: Data $x_1, \dots, x_n \sim P_\theta$, prior means $\mu_1, \dots, \mu_k$, prior $\theta \sim \Pi_\theta$ and cardinality weights $g_1, \dots, g_s$ with $\sum g = 1, (\mathbf{m} \rightarrow g(\mathbf{m}))$

Output: Partition estimate $\hat{I}$

// MAP-Estimator

```

1: Generate partition space  $\mathcal{I}_k$ 
2: for each  $l \in [k]$  do
3:   for each  $i \in [n]$  do
4:     Compute the marginal likelihood  $a_l \leftarrow (P_{X,l,i}(x_i))_{l,i}$ 
5:   end for
6: end for
7: Compute polynomial  $Z \rightarrow P[a_1, \dots, a_k](Z)$  // Lemma 2.12
8: Extract powers of monomials of P to get cardinalities space  $\mathcal{M}$ 
9: for each cardinality  $\mathbf{m} \in \mathcal{M}$  do
10:   Assign prior weight  $g_{\mathbf{m}}$  to  $\mathbf{m}$ 
11:   Extract coefficients  $A_{\mathbf{m}}$ 
12: end for
13: for partition  $I$  in  $\mathcal{I}_k$  do
14:   Compute cardinalities  $\mathbf{m} \leftarrow (|I_1|, \dots, |I_k|)$ 
15:   Compute marginal likelihood  $P_{X,I}(x) = \prod_{l \in [k]} \prod_{i \in I_l} P_{X,l,i}(x_i)$ 
16:   Compute  $\lambda_I = \binom{n}{\mathbf{m}}^{-1} g(\mathbf{m})$ 
17: end for
18: Compute  $P_X = \sum_{\mathbf{m} \in \mathcal{M}} g(\mathbf{m}) A_{\mathbf{m}}(P_{X,1}(x), P_{X,2}(x), \dots, P_{X,k}(x))$ 
19: return Partition  $\hat{I} = \arg \max_I \Pi(I|x)$ 

```

Point-wise Posterior

Remark. In the clustering setting for unknown partition where a product-form prior is assumed on I , it holds that the posterior $\Pi(I|X) = \bigotimes_{l \in [k]} \bigotimes_{i \in I_l} \Pi(i \in I_l|X)$ or, alternatively, $\Pi(I|X) = \bigotimes_{i \in [n]} \Pi(c_i|X)$ and the MAP-estimator simplifies to,

$$\hat{I}_{MAP} = \arg \max_{I \in \mathcal{I}_k} \Pi(I|X) = \arg \max_{c_1, \dots, c_n} \prod_{i \in [n]} \Pi(c_i|X)$$

where $c_1, \dots, c_n$ is an equivalent expression of partition $I \in \mathcal{I}_k$.

Definition 2.13. Let the clustering model of (2.4) for $X = (X_1 \dots X_n) \in \mathbb{R}^n$ and given the I-posterior according to the bayesian analysis. We define the point-wise posterior by,

$$p_{l^*,i} := \Pi(i \in I_{l^*} | X) = \sum_{I: i \in I_{l^*}} \Pi(I | X).$$

Remark. In case of the cardinality-based prior, the MAP estimate does *not* simplify to the component -wise posterior due to the combinatorial dependence of the cardinalities of partition I . However, the point-wise posterior can, still, be used independently to derive an estimator, namely, the Maximum Point-wise A-Posteriori Estimator, $\hat{I} = \hat{I}_{MPAP}$.

3. Product-form prior: point-wise posterior

Input: Data $x_1, \dots, x_n \sim P_\theta$, matrix $\Lambda = (\lambda_{li}) \in \mathbb{R}^{(k-1) \times N}$, prior means $\mu_1, \dots, \mu_k$ and prior $\theta \sim \Pi_\theta$

Output: Partition Estimator $\hat{I}$

// w.r.t. MPAP

- 1: Compute normalizing constants, C_λ, P_X
 - 2: **for** each $i \in [n]$ **do**
 - 3: **for** each $l \in [k]$ **do**
 - 4: Compute posterior probability $p_{l,i} = \frac{\omega_{l,i} P_{X,I_{l,i}}(x_i)}{\sum_{l=1}^k \omega_{l,i} P_{X,I_{l,i}}(x_i)}$
 - 5: **end for**
 - 6: Compute labels $c_i \leftarrow \arg \max_l p_{li}$
 - 7: **end for**
 - 8: **return** Partition $\hat{I} = (c_1, \dots, c_n)$
-

Lemma 2.14. Let sample $X = (X_1, \dots, X_n) \in \mathbb{R}^n$, $\theta \in \mathbb{L}_{\mathcal{I}_k}$ and the model as in 2.4. Suppose product-form prior $I \sim \lambda_I^{prod}$, it follows that the component-wise posterior for unknown $I \in \mathcal{I}_k$,

$$p_{l^*,i} = \sum_{I: i \in I_{l^*}} \Pi(I | X) = \frac{\omega_{l^*,i} \cdot P_{X,I_{l^*,i}}(x_i)}{\sum_{l=1}^k \omega_{l,i} P_{X,I_{l,i}}(x_i)} \quad \text{for } l^* \in [k], i \in [n].$$

Proof. See Appendix A.1. □

The implementation is described by Algorithm 3.

Lemma 2.15. Let sample $X = (X_1, \dots, X_n) \in \mathbb{R}^n$, $\theta \in \mathbb{L}_{\mathcal{I}_k}$ and the model as in 2.4. Suppose cardinality-based prior $I \sim \lambda_I^{card}$, it follows that the component-wise posterior for unknown $I \in \mathcal{I}_k$,

$$p_{l^*,i} = \frac{P_{X,I_{l^*},i}(x_i)}{P_X} \sum_{\mathbf{m} \in \mathcal{M}} g(\mathbf{m}) A_{\mathbf{m}_{l^*,-i}}(P_{X,I_1}(x), P_{X,I_2}(x), \dots, P_{X,I_k}(x)), \quad \text{for } l^* \in [k].$$

where $A_{\mathbf{m}_{l^*,-i}}$ is the $\mathbf{m}_{l^*}$ coefficient of the multivariate polynomial spanned by $n-1$ points. Refer to the following Lemma.

Proof. Given the posterior from the bayesian analysis, Theorem (2.11), it follows that,

$$\begin{aligned} p_{l^*,i} &= \sum_{I: i \in I_l} \Pi(I|X) = \sum_{I: i \in I_{l^*}} \lambda_I \frac{\bigotimes_{l \in [k]} \bigotimes_{i \in I_l} P_{X,I_l,i}(x_i)}{P_X} \\ &= \frac{1}{P_X} \sum_{\mathbf{m} \in \mathcal{M}} \sum_{\substack{I: |I|=\mathbf{m} \\ i \in I_{l^*}}} g(\mathbf{m}) \bigotimes_{l \in [k]} \bigotimes_{j \in I_l} P_{X,I_l,j}(x_j) \\ &= \frac{P_{X,I_{l^*},i}(x_i)}{P_X} \sum_{\mathbf{m} \in \mathcal{M}} \sum_{\substack{I: |I|=\mathbf{m} \\ i \in I_{l^*}}} g(\mathbf{m}) \bigotimes_{l \in [k]} \bigotimes_{\substack{j \in I_l \\ j \neq i}} P_{X,I_l,j}(x_j). \end{aligned}$$

Set $\psi_1 = P_{X,1}, \dots, \psi_k = P_{X,k}$ for $n-1$ points, with indices $[n^*] = [n] \setminus i$ and let the mult. polynomial, as given by Lemma 2.12 with $Z_1 = 1$. It follows that the obtained cardinalities,

$$\mathbf{m}_{l^*} = \begin{cases} (m_1, \dots, m_{l^*}-1, \dots, m_k), & \text{for } l^* \in [k-1] \\ \mathbf{m} & l^* = k \end{cases}$$

where $\mathbf{m} = (m_1, \dots, m_k) \in \mathcal{M}$ are the cardinalities of $I_1, \dots, I_k$ for partitions of n points into k -groups. We denote by $A_{\mathbf{m}_{l^*,-i}}$ the coefficients of the resulted multivariate polynomial of n^* points and using the Lemma 2.12, the final expression is derived.

$$p_{l^*,i} = \frac{P_{X,I_{l^*},i}(x_i)}{P_X} \sum_{\mathbf{m} \in \mathcal{M}} g(\mathbf{m}) A_{\mathbf{m}_{l^*,-i}}(P_{X,I_1}(x), P_{X,I_2}(x), \dots, P_{X,I_k}(x)), \quad \text{for } l^* \in [k].$$

□

The exact derivation of the $\hat{I}_{MPAP}$ under the cardinality-based prior assumption is given by, Algorithm 4.

4. Cardinality-based prior: Point-wise posterior

Input: Data $x_1, \dots, x_n \sim P_\theta$, prior means $\mu_1, \dots, \mu_k$, prior $\theta \sim \Pi_\theta$ and cardinality weights $g_1, \dots, g_s$ with $\sum g = 1$ ($\mathbf{m} \rightarrow g(\mathbf{m})$)

Output: Partition Estimator $\hat{I}$ // w.r.t. MPAP

```
1: for each  $l \in \{1, \dots, k\}$  do
2:   for each  $i \in \{1, \dots, n\}$  do
3:     Compute the marginal likelihood  $\alpha_{li} \leftarrow P_{X,l,i}(x_i)$ 
4:   end for
5: end for
6: Compute polynomial  $Z \rightarrow P[a_1, \dots, a_k](Z)$  // Lemma 2.12
7: Extract powers of monomials of P to get cardinalities space  $\mathcal{M}$ 
8: for each cardinality  $\mathbf{m} \in \mathcal{M}$  do
9:   Assign prior weight  $g_{\mathbf{m}}$  to  $\mathbf{m}$ 
10:  Extract coefficients  $A_{\mathbf{m}}$ 
11: end for
12: Compute  $P_X = \sum_{\mathbf{m} \in \mathcal{M}} g(\mathbf{m}) A_{\mathbf{m}}(P_{X,1}(x), P_{X,2}(x), \dots, P_{X,k}(x))$ 
13: for each  $i \in [n]$  do
14:   Compute polynomial  $Z \rightarrow P'[a_1, \dots, a_k]_{-i}(Z)$ 
15:   Extract powers of monomials of P' to get cardinalities space  $\mathcal{M}'$ 
16:   for each  $l \in [k]$  do
17:     Compute  $A_{\mathbf{m}_l, -i}(P_{X,I_1}(x), P_{X,I_2}(x), \dots, P_{X,I_k}(x))$  // Lemma 2.15
18:     Assign  $g_{\mathbf{m}_l}$  to  $\mathbf{m}_l$ 
19:     Compute  $p_{l,i} \leftarrow \frac{P_{X,I_l,i}(x_i)}{P_X} \sum_{\mathbf{m}_l \in \mathcal{M}'} g(\mathbf{m}_l) A_{\mathbf{m}_l, -i}(P_{X,I_1}(x), P_{X,I_2}(x), \dots, P_{X,I_k}(x))$ 
20:   end for
21:   Compute labels  $c_i = \arg \max_l p_{l,i}$ 
22: end for
23: return Partition  $\hat{I} = (c_1, \dots, c_n)$ 
```

Remark. The computation of $\hat{I}_{\text{MPAP}}$ under the cardinality-based prior requires the generation of $n + 1$ polynomials still involve $\mathcal{O}(k^n)$ coefficients. Thus, Algorithm 4 remains computationally infeasible.

2.7 Example: Normal Data

In order to simplify the computational complexity already, we take advantage of conjugacy by assuming that the data and the prior on θ follow Normal distributions, in

the following way,

$$\begin{aligned} X_i &\sim \mathcal{N}(\theta_i, 1), \quad \forall i \in [n] \\ \theta_i | I &\sim \Pi_{I,i} = \mathcal{N}(\mu_l, 1), \quad i \in I_l \\ I &\sim \lambda_I \end{aligned}$$

where $\mu_1, \dots, \mu_l$ are the prior means of θ . In addition, it holds that $\theta \sim \sum_{I \in \mathcal{I}_k} \lambda_I \Pi_I$ which in the product-form prior case is rewritten as, $\theta_i \sim \sum_{l=1}^k \omega_{l,i} \Pi_{l,i}, \forall i \in [n]$. Using the conjugacy of the posterior of $\theta_i \in I_l$ for each group I_l is,

$$\theta_i | X_{I_l}, I_l \sim \mathcal{N}\left(\frac{\sum_{i \in I_l} X_i + \mu_l}{n_l + 1}, \frac{1}{n_l + 1}\right).$$

2.8 Multidimensional Extension

Up to this point, we have assumed univariate observations, i.e., $X_1, \dots, X_n \in \mathbb{R}$. We now extend the framework to the multivariate setting.

Remark. In the multi-dimensional setting, $X = (X_1, \dots, X_n) \in \mathbb{R}^{n \times d}$, where d denotes the dimensionality of the space. The Bayesian clustering construction remains invariant to changes in d , under the model described in 2.4.

Remark. Given model 2.4 and a data point $x \in \mathbb{R}^d$, the computation of the posterior density $P_{X,I,i}(x)$ becomes increasingly complex with higher d , as it requires d -dimensional integration:

$$P_{X,I,i}(x) = \int \cdot^d \cdot \int P_{\theta_i}(x) d\Pi_{I,i}(\theta_i)$$

To simplify analysis and facilitate visualization, we focus on the 2-dimensional case $X_1, \dots, X_n \in \mathbb{R}^2$. Consequently, the likelihood P_θ and the prior Π are bivariate.

2.9 Distance Measures

To provide with a comprehensive assessment of the clustering quality, we employ three distance measures between the true clustering $\mathbf{c}$ and the estimated clustering $\hat{\mathbf{c}}$:

Adjusted Rand Index (ARI) addressing statistical agreement, *Variation of Information (VI)* as a loss function and the *Perfect Match Distance (PMD)* for exact label correspondence.

- **Adjusted Rand Index (ARI)** is a chance-adjusted version of the Rand Index, which measures the agreement between two clusterings by comparing all pairs of points and determining whether they are assigned consistently. ARI is defined as:

$$\text{ARI} = \frac{\text{RI} - \mathbb{E}[\text{RI}]}{\max(\text{RI}) - \mathbb{E}[\text{RI}]}$$

where RI is the Rand Index and $\mathbb{E}[\text{RI}]$ its expected value under a random model. Values of ARI close to 1 indicate high agreement, while values near 0 suggest randomness.

- **Variation of Information (VI)**, introduced by Meilă (2007) [10], is an information-theoretic metric defined as:

$$\text{VI}(\mathbf{c}, \hat{\mathbf{c}}) = H(\mathbf{c}) + H(\hat{\mathbf{c}}) - 2I(\mathbf{c}; \hat{\mathbf{c}})$$

where $H(\cdot)$ denotes the entropy of a clustering, and $I(\cdot; \cdot)$ is the mutual information between two clusterings. We report a normalized VI score, adjusted by dividing by $2\log K^*$, where K^* is the maximum number of clusters in either $\mathbf{c}$ or $\hat{\mathbf{c}}$. Lower values of VI indicate greater similarity. We reverse this effect by using as a performance score the Adj. VI = $1 - \text{VI}/2\log(K^*)$.

- The **Perfect Match Distance (PMD)** is a combinatorial measure based on the Hungarian algorithm. It computes the minimum number of label assignments that need to be changed in $\hat{\mathbf{c}}$ to exactly match $\mathbf{c}$. Formally, this is expressed as:

$$\text{PMD} = \min_{\sigma \in S_K} \sum_{i=1}^N \mathbf{1}\{\mathbf{c}_i \neq \sigma(\hat{\mathbf{c}}_i)\}$$

where σ is a permutation over cluster labels and S_K the symmetric group over K elements. PMD provides an intuitive count of mismatched observations. We define a performance metric, with values close to 1 indicating similarity and values close to 0 indicating dissimilarity, as follows, Adj. PMD = $1 - \text{PMD}/N$.

3 Simulation study

3.1 Simulated Examples

Multiple datasets are generated, each of size n as mentioned in Table 3.1, from k bivariate normal distributions. Given a dataset of size n and k clusters, each data point $X_{i,l}$ is drawn i.i.d. as follows,

$$X_{i,l} \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}\left(\begin{bmatrix} \mu_{l,1} \\ \mu_{l,2} \end{bmatrix}^T, \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\right),$$

where $i \in I_k$ and $\mu_l = (\mu_{l,1}, \mu_{l,2})$ for $l \in [k]$ are the cluster means which are positioned with a radius proportional to δ around the middle fixed cluster mean $\mu_{\lfloor \frac{k}{2} \rfloor}$, which is set to $[1, 1]$ in the following way,

$$(\mu_{l,1} - 1)^2 + (\mu_{l,2} - 1)^2 = (r * \delta)^2 \quad \text{and} \quad r = |l - \lfloor \frac{k}{2} \rfloor|.$$

Remark. Since the covariance matrix is set to be the identity, the parameter δ controls the distance between the cluster means μ .

Varying the parameters, given in Table 3, allows for the evaluation of algorithm performance across a spectrum of clustering scenarios.

Parameters

- The **Sample sizes** n range from very small (20) to large-scale settings (1000), enabling us to observe performance under both data scarcity and abundance.
- The **number of clusters**, k , varies from small (3), moderate (7), to large (11) numbers of clusters, introducing varying levels of noise to the synthetic data.
- The **δ -parameter** ranges from very small (0.2), where clusters are highly overlapping, to very large (6.0), where clusters are well-separated. This range enables us to

evaluate the algorithm’s robustness to noise and its ability to resolve subtle cluster structures. The relation between δ and covariance matrix plays a significant role in shaping the data structure.

Parameter	Values
Dimensions	2
Clusters	3, 5, 7, 11
Delta	0.2, 0.5, 0.8, 1.1, 2.5, 6.0
Sample Size	20, 50, 80, 100, 200, 500, 800, 1000

Table 3.1: Simulation design: parameters and their respective values.

3.2 Results

Number of Clusters

As shown in Figures 3.1 and 3.2, the number of clusters significantly influences the performance of the ProdForm algorithm. When clusters overlap (low δ), as in the case of $k = 3$, the algorithm performs marginally better than for larger values of k , likely due to reduced noise and simpler structure as opposed to when clusters are well-separated (high δ), performance tends to decrease. This is likely because larger n implies fewer points per cluster and a higher presence of outliers, which counteracts to the separation provided by δ . Moreover, the ratio of sample size to number of

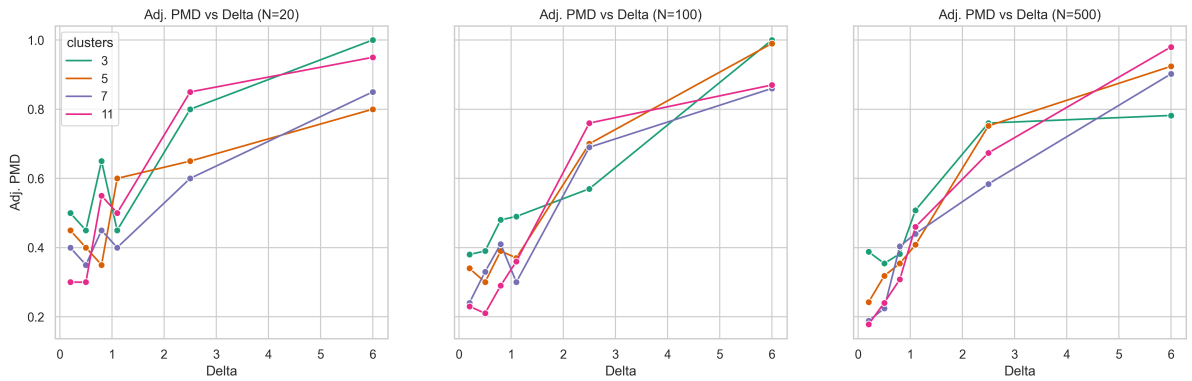


Figure 3.1: Product-form performance measured by Adj. PMD dependent vs delta distances between the means.

clusters plays a key role: for large datasets, $n = 1000$, performance improves with

larger k values, as shown in Figure 3.2. This suggests that with sufficient data, the algorithm is better able to manage the added complexity of more clusters.

Interestingly, at very small sample sizes, $n = 20$, both $k = 3$ and $k = 11$ yield better results under the Adj. PMD, when $\delta = 6$. This may be due to extreme configurations—either very dense or very sparse clusters—producing favorable alignments, possibly linked to the symmetry in the binomial distribution of cluster assignments.

Delta Effect

Again in Figures 3.1 and 3.2, for smaller values of δ , the algorithm performs worse under both the Adj. VI and Adj. PMD, due to overlapping clusters. The upward trend with δ is evident reaching performance scores close to 1 for $\delta = 6$ for all sample sizes. Given small δ , for smaller sample sizes, there are not enough points to capture the shape of the clusters and therefore ProdForm struggles to capture the true partition leading to higher variability in the results. As the sample size increases, the performance trend becomes clearer, less variable, with the best Adj. PMD scores observed when the clusters are well-separated.

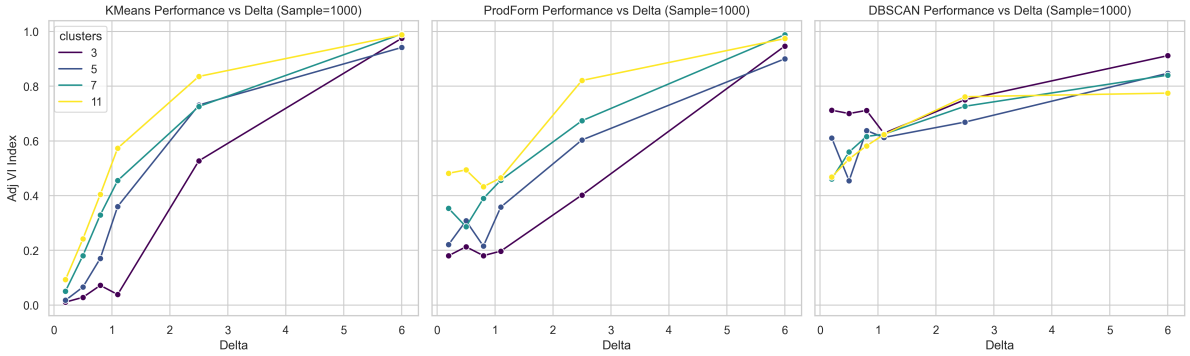


Figure 3.2: Performance measured by Adj. VI of Kmeans, ProdForm, DBScan with a sample size of $N = 1000$.

Comparison with KMeans & DBScan

In Figure 3.2, for $N = 1000$, the performance of the product-form method follows a similar trend to KMeans. When the clusters are overlapping, the product-form algorithm tends to outperform KMeans. This is likely because KMeans struggles with overlapping clusters, while the product-form method incorporates uncertainty into

the model, leading to better results. Compared to DBScan, the product-form method is more sensitive to overlapping clusters, whereas DBScan's performance is more stable across different cluster configurations.

The performance of the methods with respect to sample size, shown in Figure 3.3, follows an overall upward trend as δ increases. For small values of δ , product-form performs better than KMeans and worse than DBScan. When the clusters are sufficiently far apart, the performance of all methods stabilizes, with product-form achieving around 0.7 in terms of the Rand index. Finally, when the clusters are well-separated, the performance is around 0.9 for the ProdForm algorithm reaching its maximum performance for $n = 1000$.

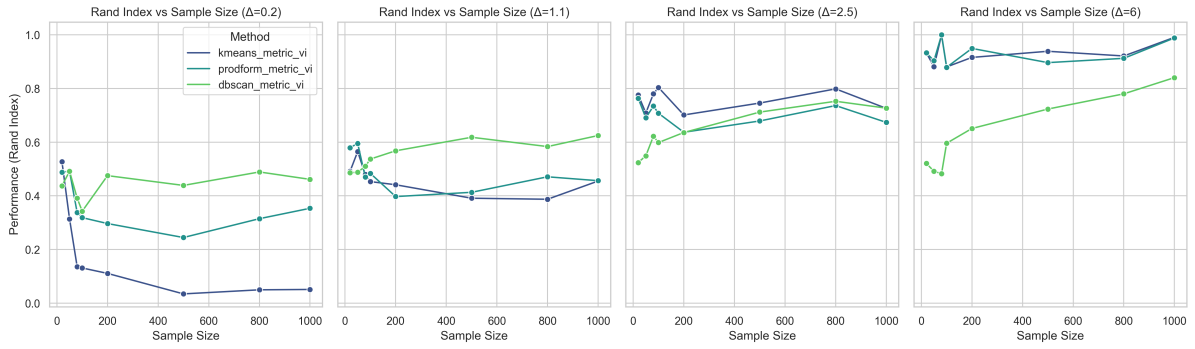


Figure 3.3: Performance measured by ARI of Kmeans, ProdForm, DBScan as a function of sample size.

4 Real data and application

4.1 The iris dataset

The Iris dataset is a widely used benchmark dataset in pattern recognition, consisting of 150 samples from three species of Iris flowers (Setosa, Versicolor, and Virginica). Each sample is described by four numerical features: sepal length, sepal width, petal length, and petal width (all in centimeters). The objective is to classify the species based on these measurements.

Feature Aggregation

To enable clustering and visualization in a two-dimensional space, we derive two composite features by computing the approximate areas of the sepals and petals:

$$\text{sepal area} = \text{sepal length} \times \text{sepal width},$$

$$\text{petal area} = \text{petal length} \times \text{petal width}.$$

These features serve as rough proxies for the physical areas of the flower parts. Each flower sample is then represented as a two-dimensional vector:

$$\mathbf{flower}_i = \begin{bmatrix} \text{petal area}_i \\ \text{sepal area}_i \end{bmatrix}$$

This transformation reduces the original 4-dimensional feature space to a two-dimensional representation by combining sepal and petal dimensions into area-based features. This simplification facilitates visual interpretation and enables the application of unsupervised clustering techniques such as K-Means and DBSCAN.

4.2 Results

Prior mean selection

A key observation when applying the product-form clustering algorithm is the strong dependence of the resulting cluster shapes on the initial choice of the prior means, μ_θ . For example, when the prior means are initialized along a horizontal line in the feature space, the resulting clusters tend to exhibit a horizontally stretched structure. Conversely, initializing the means along a vertical line produces vertically oriented clusters. As show in Figure 4.1, when prior means are selected as random data points,

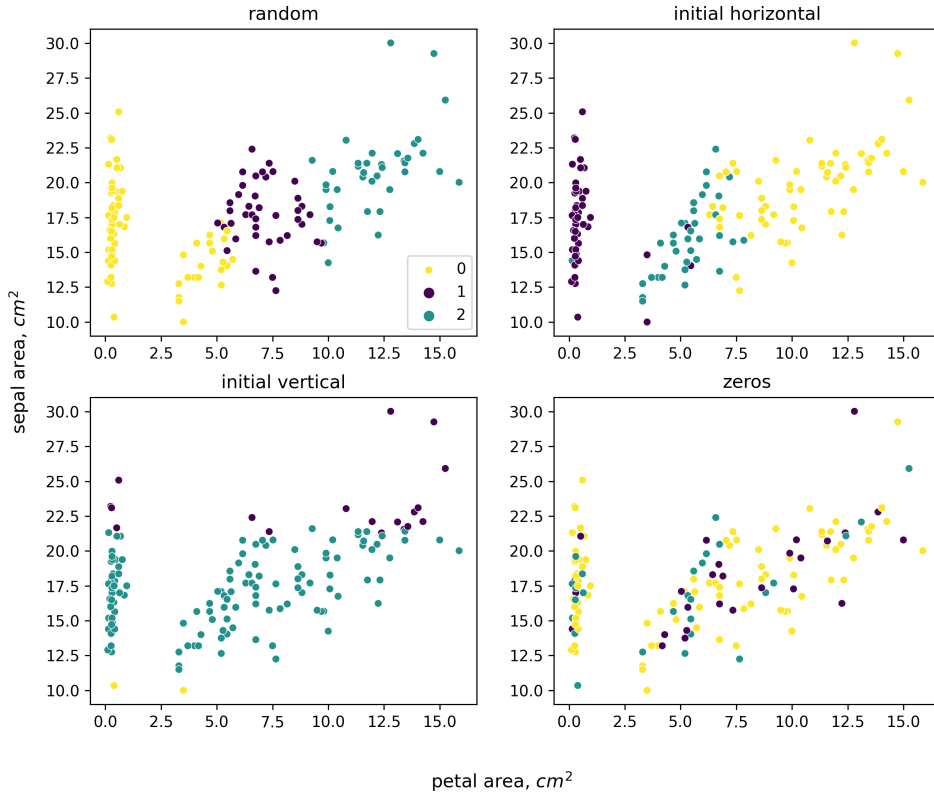


Figure 4.1: Product-form clustering with initialization of prior means, assuming 3-clusters.

the results can vary significantly. In some cases, this leads to overlapping or poorly separated clusters, particularly when the sampled means are not representative of the data distribution. Similarly, initializing all prior means at the origin fails to provide meaningful partitioning, as the clustering outcome lacks structure and interpretability. These findings underscore the importance of informed initialization in

Bayesian clustering models, particularly in low-dimensional representations of complex datasets.

In order to provide with meaningful insights, in the following analysis we use as prior means the centers of the resulted k-means algorithm giving rise to adaption to Empirical Bayes and we compare the clusterings provided by KMeans with our approach.

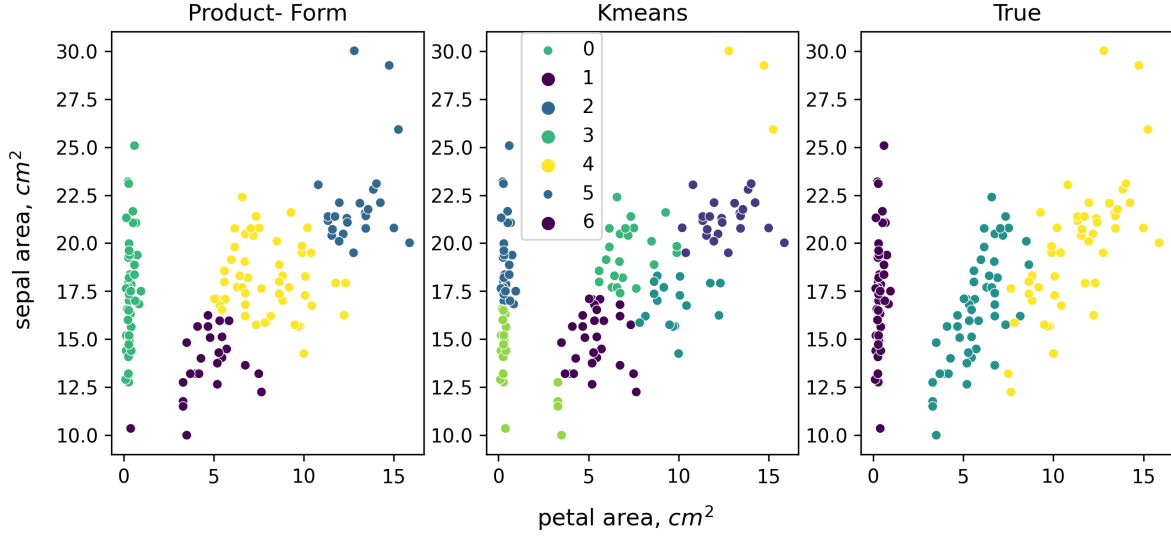


Figure 4.2: Clustering of Iris dataset samples based on petal and sepal area.

Number of clusters

As illustrated in Figure 4.2, initializing the model with $k = 7$ clusters serves as an upper bound on the number of clusters that can emerge. While traditional methods such as K-Means impose this cluster count as a fixed constraint, the Bayesian clustering approach allows for a more flexible structure. It infers the optimal number of clusters dynamically based on the estimated posterior partition index, $\hat{I}_{\text{MPAP}}$.

4.2.1 Model assumptions

We assume that each observation follows a normal model, $X_i \sim \mathcal{N}(\theta_i, \mathbf{I}_d)$. However, due to the unsupervised nature of the clustering process, this assumption cannot be verified prior to the analysis. As a result, standard normalization techniques are not effective and, thus, model misspecification is unavoidable. Once the partition esti-

mate $\hat{I}_{\text{MPAP}}$ has been obtained, the validity of the underlying distributional assumptions can be assessed retrospectively,

$$\forall i \in [n], \quad \theta_i | \hat{I} \sim \mathcal{N}(\mu_i, \mathbf{1}), \quad \mu_i = \mu_l \text{ for } i \in \hat{I}_l.$$

and

$$\forall i \in [n], \quad X_i | \hat{I} \sim \mathcal{N}(\theta_i, \mathbf{1}), \quad \theta_i = \theta_l \text{ for } i \in \hat{I}_l.$$

Covariance

The first step is to check the covariance between `sepal_area` and `petal_area`. The Pearson correlation test fails to reject the null hypothesis, indicating that there is no significant correlation between the two variables. Therefore, the bivariate normal distribution can be considered as the product of two univariate distributions.

Cluster	Pearson r	p-value	Significant
0	0.2607	0.2296	No
1	0.3097	0.1409	No
2	0.2836	0.0534	No
3	0.1461	0.2827	No

Table 4.1: Pearson Correlation Results by Cluster

Normality

To assess the normality of the data, standard techniques such as the Shapiro–Wilk test, the Kolmogorov–Smirnov (KS) test, and Q–Q plots were performed. Overall, the assumption of normality appears reasonable across clusters. However, special attention should be given to the petal area distribution in Cluster 2, where a deviation from normality is indicated. A summary of the KS test results is provided in Table 4.2, while additional diagnostic plots and test outcomes are available in the Appendix A.2.

Cluster	Feature	KS-statistic	p-value	Normal ($p > 0.00625$)
0	Sepal Area	0.0945	0.9741	Yes
1	Sepal Area	0.2492	0.0846	Yes
2	Sepal Area	0.0866	0.8426	Yes
3	Sepal Area	0.1019	0.5710	Yes
0	Petal Area	0.1911	0.3275	Yes
1	Petal Area	0.0727	0.9985	Yes
2	Petal Area	0.2660	0.0020	No
3	Petal Area	0.1214	0.3524	Yes

Table 4.2: Kolmogorov–Smirnov test results for normality of sepal and petal area distributions within each cluster. Normality is assessed at a Bonferroni-adjusted threshold of $\alpha = 0.00625$.

4.2.2 Distribution of Clusters

We consider each cluster individually under a two-step Hierarchical Bayesian framework. For a fixed cluster $l \in [k^*] \subseteq \{1, \dots, k\}$, the following model is assumed:

$$\theta \mid \hat{I} \sim \mathcal{N}(\mu_l, \mathbf{1}),$$

$$X \mid \hat{I} \sim \mathcal{N}(\theta_l, \mathbf{1}),$$

where $\hat{I}$ denotes the estimated partition under the MPAP. Given the conjugate nature of the model, the posterior mean μ_l is analytically tractable.

To validate the assumed variance, we perform a chi-squared test for each cluster. The null hypothesis that the cluster variance is equal to one is rejected in all cases. As a result, the sample variance is used as an empirical estimate of the true variance within each cluster. The resulting fitted distributions per cluster are illustrated below.

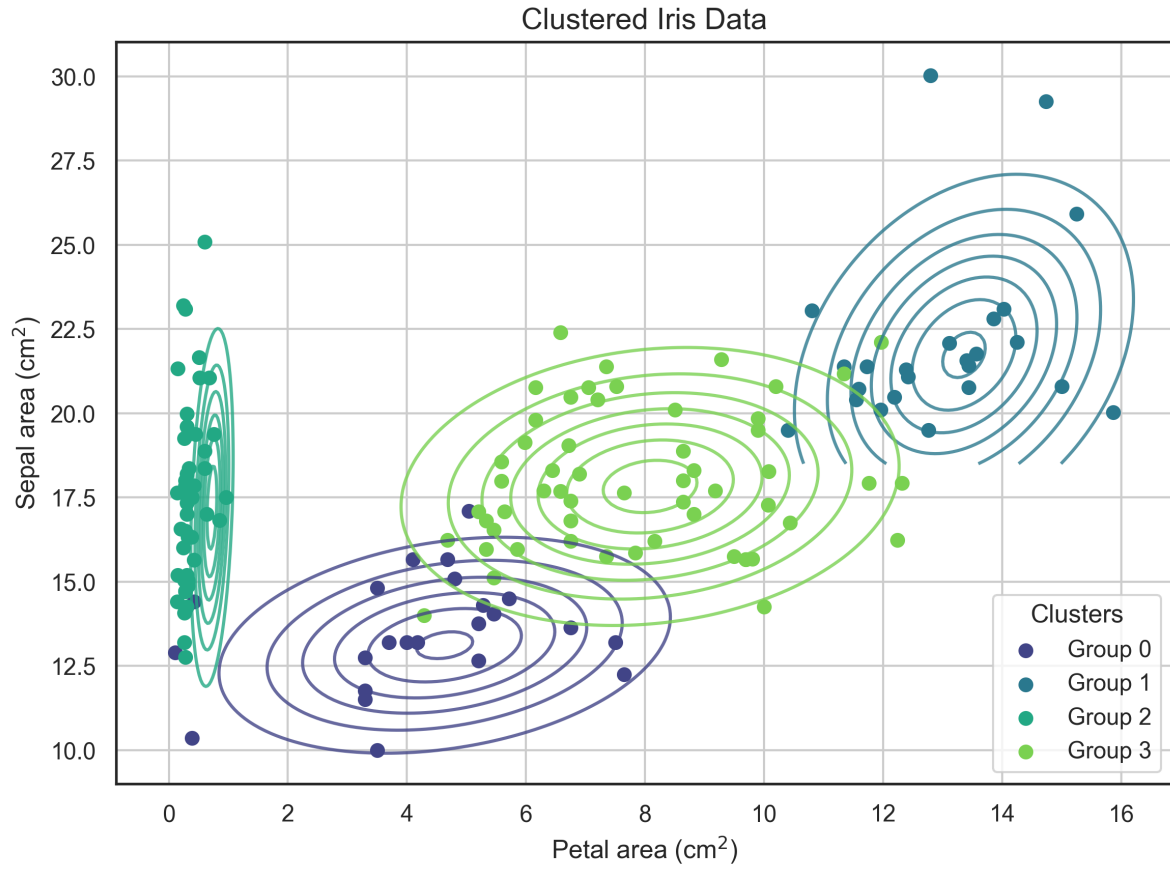


Figure 4.3: Retrieval of Distribution of Iris Data for each clusters provided the $\hat{I}_{MPAP}$ -estimate using a Product-Form prior.

5 Conclusion

The selection of prior under the signal plus noise model has been address for clustering general k-groups purposes. The cardinality-based prior lacks scalability and thus is not applicable in real-world data. The Product-Form clustering approach exhibits strong empirical performance across both synthetic and real datasets as it displays performance similar to methods cutting edge methods such as *Kmeans*.

The probabilistic foundation of the ProdForm method offers a principled way to account for uncertainty in cluster assignment. A key advantage of this approach is its ability to specify an upper bound on the number of clusters, which improves flexibility and interpretability. Nonetheless, the method remains sensitive to the initialization of prior means and relies on certain assumptions regarding the distribution of cluster components.

Future Work

Future directions include the development of adaptive strategies for selecting prior means, potentially through empirical Bayes methods or data-driven initialization schemes. Improving the scalability and efficiency of posterior inference—particularly in high-dimensional or non-Gaussian settings—is another promising avenue. While the current framework is limited by the assumption of normality, incorporating analytical tractability for alternative distributions could broaden its applicability to more complex clustering scenarios. Addressing these challenges would enhance the robustness and versatility of the ProdForm clustering method in practical applications.

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Appendix

A.1 Proofs - extended formulas

Clustering in 2-groups

Proof of Lemma 2.2. Considering the mixture assumptions on θ , the marginal likelihood will be rewritten as,

$$\begin{aligned} P_X &= \sum_{I \in \mathcal{I}_k} \lambda_I P_{X,I} = \sum_{I \in \mathcal{I}_k} \lambda_I \bigotimes_{i \in [n]} P_{X,I,i}(x_i) \\ &= \sum_{I \in \mathcal{I}_k} \lambda_I \bigotimes_{l \in [k]} \bigotimes_{i \in I_l} P_{X,I_l,i}(x_i). \end{aligned}$$

Similarly, the posterior, $\Pi(I|X)$,

$$\begin{aligned} \Pi(I|X) &= \frac{\lambda_I P_{X,I}}{P_X} = \frac{\lambda_I}{P_X} \bigotimes_{i \in [n]} P_{X,I,i}(x_i) \\ &= \frac{\lambda_I}{P_X} \bigotimes_{l \in [k]} \bigotimes_{i \in I_l} P_{X,I_l,i}(x_i). \end{aligned}$$

The posterior $\Pi(\theta|X)$,

$$\begin{aligned}
\Pi(\theta|X) &= \sum_{I \in \mathcal{I}_k} \Pi_I(\theta|X) \Pi(I|X) \\
&= \sum_{I \in \mathcal{I}_k} \lambda_I \bigotimes_{l \in [k]} \bigotimes_{i \in I_l} \Pi_{I_l, i}(\theta_i|X_i) \frac{\bigotimes_{l \in [k]} \bigotimes_{i \in I_l} P_{X, I_l, i}(x_i)}{P_X} \\
&= \sum_{I \in \mathcal{I}_k} \frac{\lambda_I}{P_X} \bigotimes_{l \in [k]} \bigotimes_{i \in I_l} \Pi_{I_l, i}(\theta_i|X_i) \times P_{X, I_l, i}(x_i).
\end{aligned}$$

Using $\Pi_{I_l, i}(\theta_i|X_i) = \frac{\Pi_{I_l, i}(\theta_i) \cdot P_{\theta_i}(X_i)}{P_{X, I_l, i}(X_i)}$, for $i \in [n]$, $l \in [k]$, we arrive at the final expression,

$$\Pi(\theta|X) = \sum_{I \in \mathcal{I}_k} \lambda_I \frac{\bigotimes_{l \in [k]} \bigotimes_{i \in I_l} \Pi_{I_l, i}(\theta_i) \cdot P_{\theta_i}(x_i)}{P_X}.$$

□

Proposition A.1 (Construction of C_λ). Assume prior λ on partition space $\mathcal{I}_2$, $I \sim \lambda_I$, and $I = (I_o, I_o^c) \in \mathcal{I}_2$ such that, $\lambda_I = C_\lambda \prod_{i \in I_o} \lambda_i$ where λ_i are the prior weights $\forall i \in [n]$. It must hold that, $C_\lambda = \prod_{i=1}^n (1 + \lambda_i)^{-1}$.

Proof. Suppose λ a prior on $\mathcal{I}_2$ s.t. $\lambda_I = C_\lambda \prod_{i \in I_o} \lambda_i$. It must hold that, $\sum_{I \in \mathcal{I}_2} \lambda_I = 1$. Combining the two expressions results,

$$\begin{aligned}
\sum_{I \in \mathcal{I}_2} \lambda_I &= \sum_{I \in \mathcal{I}_2} C_\lambda \prod_{i \in I_o} \lambda_i \\
&= \sum_{I \in \mathcal{I}_2} C_\lambda \prod_{i \in I_o} \lambda_i \prod_{j \in I_o^c} 1.
\end{aligned}$$

By using the following elementary identity for $a_i = \lambda_i$, and $b_i = 1$, $\forall i \in [n]$,

$$\prod_{i \in [n]} (a_i + b_i) = \sum_{I \in \mathcal{I}_2} \left(\prod_{i \in I_o} a_i \right) \left(\prod_{j \in I_o^c} b_j \right) \text{ for } b_j = 1, \forall j \in [n]. \quad (\text{A.1})$$

we arrive at the following expression, $\sum_{I \in \mathcal{I}_2} \lambda_I = C_\lambda \prod_{i \in [n]} (\lambda_i + 1)$. Thus, by requiring

$C_\lambda = \prod_{i \in [n]} (\lambda_i + 1)^{-1}$, the prior assumption of λ_I holds. Consequently,

$$\lambda_I = C_\lambda \prod_{i \in I_0} \lambda_i, \quad C_\lambda = \prod_{i \in [n]} (\lambda_i + 1)^{-1}.$$

□

Proof of Theorem 2.4. By substituting the product-form prior $I \sim \lambda_I^{prod}$ into the given expressions, the following holds,

$$\begin{aligned} \Pi &= \sum_{I \in \mathcal{I}_2} \lambda_I^{prod} \Pi_I = \sum_{I \in \mathcal{I}_2} C_\lambda \prod_{i \in I_0} \lambda_i \left[\bigotimes_{i \in I_0} \Pi_{1,i} \times \bigotimes_{j \in I_0^c} \Pi_{2,j} \right] \\ &= C_\lambda \sum_{I \in \mathcal{I}} \left(\bigotimes_{i \in I_0} \lambda_i \Pi_{1,i} \times \bigotimes_{j \in I_0^c} \Pi_{2,j} \right) \end{aligned}$$

By using (A.3) and letting $a_i = \Pi_{1,i}$ and $b_i = \Pi_{2,i}$, $\forall i \in [n]$,

$$\begin{aligned} \Pi &= \prod_{i \in [n]} (1 + \lambda_i)^{-1} \bigotimes_{i \in [n]} \left[\lambda_i \Pi_{1,i} + \Pi_{2,i} \right] \\ &= \bigotimes_{i \in [n]} \left[\lambda_i (\lambda_i + 1)^{-1} \Pi_{1,i} + (1 + \lambda_i)^{-1} \Pi_{2,i} \right]. \end{aligned}$$

Finally, $\Pi = \bigotimes_{i \in [n]} \left[\omega_i \Pi_{1,i} + (1 - \omega_i) \Pi_{2,i} \right]$, with $\omega_i = \frac{\lambda_i}{1 + \lambda_i}$, $i \in [n]$.

Similarly for the marginal,

$$\begin{aligned} P_X &= \sum_{I \in \mathcal{I}_2} \lambda_I P_{X,I}(x) = \sum_{I \in \mathcal{I}_2} \left(C_\lambda \prod_{i \in I_0} \lambda_i \bigotimes_{i \in I_0} P_{X,1,i}(x_i) \times \bigotimes_{j \in I_0^c} P_{X,2,j}(x_j) \right) \\ &= C_\lambda \sum_{I \in \mathcal{I}} \left(\bigotimes_{i \in I_0} \lambda_i P_{X,1,i}(x_i) \times \bigotimes_{j \in I_0^c} P_{X,2,j}(x_j) \right) \\ &\stackrel{(A.3)}{=} C_\lambda \bigotimes_{i \in [n]} \left[\lambda_i P_{X,1,i}(x_i) + P_{X,2,i}(x_i) \right] \\ &= \bigotimes_{i \in [n]} \left[\omega_i P_{X,1,i}(x_i) + (1 - \omega_i) P_{X,2,i}(x_i) \right]. \end{aligned}$$

The I-posterior $\Pi(I|X)$,

$$\begin{aligned}
\Pi(I|X) &= \frac{\lambda_I P_{X,I}}{P_X} = \frac{C_\lambda \prod_{i \in I_o} \lambda_i P_{X,I}}{P_X} \\
&= \frac{C_\lambda}{P_X} \prod_{i \in I_o} \lambda_i \bigotimes_{i \in I_o} P_{X,1,i}(x_i) \times \bigotimes_{j \in I_o^c} P_{X,2,j}(x_j) \\
&= \frac{C_\lambda}{P_X} \bigotimes_{i \in I_o} \lambda_i P_{X,1,i}(x_i) \times \bigotimes_{j \in I_o^c} P_{X,2,j}(x_j).
\end{aligned}$$

Lastly, the θ -posterior,

$$\begin{aligned}
\Pi(\theta|X) &= \sum_{I \in \mathcal{I}} \Pi_I(\theta|X) \Pi(I|X) \\
&= \frac{C_\lambda}{P_X} \sum_{I \in \mathcal{I}} \left(\bigotimes_{i \in I_o} \lambda_i \Pi_{1,i}(\theta_i) P_{\theta_i}(x_i) \times \bigotimes_{j \in I_o^c} \Pi_{2,j}(\theta_j) P_{\theta_j}(x_j) \right) \\
&\stackrel{(A.3)}{=} \frac{1}{P_X} \bigotimes_{i \in [n]} [\omega_i \Pi_{1,i}(\theta_i) P_{\theta_i}(x_i) + (1 - \omega_i) \Pi_{2,i}(\theta_i) P_{\theta_i}(x_i)].
\end{aligned}$$

□

Proposition A.2 (Construction of cardinality-based prior). Assume prior λ on the partition space $\mathcal{I}_2$, $I \sim \lambda_I$ of the form $\lambda_I = c_m g(m)$ where $m \in [n]$ is the cardinality of I_o , $I = (I_o, I_o^c)$ and g nonnegative function s.t. $\sum_{m=0}^n g(m) = 1$. It must hold that $c_m = \binom{n}{m}^{-1}$.

Proof. Assuming λ_I is a prior, then $\sum_{I \in \mathcal{I}_2} \lambda_I = 1$.

$$\sum_{I \in \mathcal{I}_2} \lambda_I = \sum_{m=0}^n \sum_{I: |I|=m} c_m g(m) = \sum_{m=0}^n \binom{n}{m} c_m g(m) \tag{A.2}$$

Thus, the required m -dependent constant is, $c_m = \binom{n}{m}^{-1}$, $\forall m \in [n]_0$. □

Proof of the Polynomial Trick 2.7. The trick is used by Castillo and AW van der Vaart, [7] for implementation purposes.

Let $a = (a_1, \dots, a_n) \in \mathbb{R}^n$ and $b = (b_1, \dots, b_n) \in \mathbb{R}^n$ be vectors with components defined by

$$a_i = \psi(x_i), \quad b_i = \phi(x_i), \quad \text{for } i = 1, \dots, n,$$

where $\psi, \phi \in C^2(\mathbb{R})$ are evaluated at points $x_1, \dots, x_n \in \mathbb{R}$. Define the polynomial $P_Z(a, b) \in \mathbb{R}[Z]$ as,

$$P_Z(a, b) = \Phi(a, b) = \prod_{i=1}^n (a_i + b_i Z).$$

Its m^{th} -order coefficient, A_m is the coefficient of its m -order monomial, Z^m . Choose exactly m indices, $i \in S \subseteq [n]_0$, it follows that $|S| = m$ and

$$\prod_{i \in S} (b_i \cdot Z) \prod_{i \in S^c} a_i = \left(\prod_{i \in S} b_i \prod_{i \in S^c} a_i \right) Z^{|S|}.$$

Summing over all subsets $S \subseteq [n]_0$, it holds that $S_p = (S, S^c)$ is a partition of $[n]$ indices into 2-groups, $S_p \in \mathcal{I}_2$,

$$A_m(a, b) = \sum_{|S|=m} \left(\prod_{i \in S} b_i \prod_{i \in S^c} a_i \right).$$

□

General k-groups: product-form prior

Proposition A.3 (Construction of C_λ). Assume prior λ on partition space $\mathcal{I}_k$, $I \sim \lambda_I$, and $I = (I_1, \dots, I_k) \in \mathcal{I}_k$ such that,

$$\lambda_I^{prod} = C_\lambda \prod_{i \in I_1} \lambda_{1,i} \cdots \prod_{i \in I_{k-1}} \lambda_{k-1,i},$$

where $\lambda_{l,i}$ are the prior weights $\forall i \in [n], l \in [k]$. It must hold that,

$$C_\lambda := \prod_{i=1}^n (\lambda_{1,i} + \cdots + \lambda_{k-1,i} + 1)^{-1}.$$

Proof. Suppose λ a prior on $\mathcal{I}_k$ s.t. $\lambda_I = C_\lambda \prod_{i \in I_1} \lambda_{1,i} \cdots \prod_{i \in I_{k-1}} \lambda_{k-1,i}$ It must hold that, $\sum_{I \in \mathcal{I}_k} \lambda_I = 1$. Combining the two expressions results,

$$\begin{aligned} \sum_{I \in \mathcal{I}_k} \lambda_I &= \sum_{I \in \mathcal{I}_k} C_\lambda \prod_{i \in I_1} \lambda_{1,i} \cdots \prod_{i \in I_{k-1}} \lambda_{k-1,i} \\ &= \sum_{I \in \mathcal{I}_2} C_\lambda \prod_{i \in I_1} \lambda_{1,i} \cdots \prod_{i \in I_{k-1}} \lambda_{k-1,i} \prod_{j \in I_k} 1. \end{aligned}$$

By using the following elementary identity for $a_{l,i} = \lambda_{l,i}$ and $a_{k,i} = 1$, $\forall i \in [n], \forall l \in [k]$,

$$\prod_{i \in [n]} (a_{1,i} + \dots + a_{k,i}) = \sum_{I \in \mathcal{I}_k} \left(\prod_{i \in I_1} a_{1,i} \right) \dots \left(\prod_{i \in I_k} a_{k,i} \right), \quad (\text{A.3})$$

we arrive at the expression,

$$\sum_{I \in \mathcal{I}_k} \lambda_I = C_\lambda \sum_{I \in \mathcal{I}_k} \left(\prod_{i \in I_1} \lambda_{1,i} \right) \dots \left(\prod_{i \in I_{k-1}} \lambda_{k-1,i} \right) \left(\prod_{i \in I_k} 1 \right).$$

Thus, by requiring $C_\lambda := \prod_{i=1}^n (\lambda_{1,i} + \dots + \lambda_{k-1,i} + 1)^{-1}$, the prior assumption of λ_I holds. Consequently,

$$\lambda_I^{prod} = C_\lambda \prod_{l \in [k]} \prod_{i \in [n]} \lambda_{l,i}, \text{ where, } C_\lambda := \prod_{i=1}^n (\lambda_{1,i} + \dots + \lambda_{k-1,i} + 1)^{-1}.$$

□

Proof of Theorem 2.9. Assume $X = (X_1, \dots, X_n) \in \mathbb{R}^n$ and $I \sim \lambda_I^{prod}$. The bayesian components are rewritten as follows. The marginal likelihood,

$$\begin{aligned} P_X &= \sum_{I \in \mathcal{I}_k} \left(C_\lambda \cdot \lambda_I \bigotimes_{l \in [k]} \bigotimes_{i \in I_l} P_{X,I_l,i}(x_i) \right) = \sum_{I \in \mathcal{I}_k} \left(C_\lambda \prod_{l \in [k-1]} \prod_{i \in I_l} \lambda_{l,i} \bigotimes_{l \in [k]} \bigotimes_{i \in I_l} P_{X,I_l,i}(x_i) \right) \\ &= C_\lambda \sum_{I \in \mathcal{I}_k} \left(\bigotimes_{l \in [k-1]} \bigotimes_{i \in I_l} \lambda_{l,i} P_{X,I_l,i}(x_i) \times \bigotimes_{j \in I_k} P_{X,I_k,j}(x_j) \right) \\ &= C_\lambda \bigotimes_{i \in [n]} \left[\sum_{l \in [k-1]} \lambda_{l,i} P_{X,I_l,i}(x_i) + P_{X,I_k,i}(x_i) \right] \\ &= \bigotimes_{i \in [n]} \left[\sum_{l \in [k-1]} \omega_{l,i} P_{X,I_l,i}(x_i) + \left(1 - \sum_{l \in [k-1]} \omega_{l,i} \right) P_{X,I_k,i}(x_i) \right] \end{aligned}$$

$$\text{where } \omega_{l,i} = \frac{\lambda_{l,i}}{\sum_{s=1}^{k-1} \lambda_{s,i} + 1}, \quad l \in [k-1], i \in [n].$$

The prior,

$$\begin{aligned}
\Pi &= C_\lambda \sum_{I \in \mathcal{I}_k} \lambda_I \bigotimes_{l \in [k]} \bigotimes_{i \in I_l} \Pi_{I_l, i} = C_\lambda \sum_{I \in \mathcal{I}_k} \left(\bigotimes_{l \in [k-1]} \bigotimes_{i \in I_l} \lambda_{l, i} \Pi_{I_l, i}(x_i) \times \bigotimes_{j \in I_k} \Pi_{I_k, j}(x_j) \right) \\
&= C_\lambda \bigotimes_{i \in [n]} \left[\sum_{l \in [k-1]} \lambda_{l, i} \Pi_{I_l, i}(x_i) + \Pi_{I_k, i}(x_i) \right] \\
&= \bigotimes_{i \in [n]} \left[\sum_{l \in [k-1]} \omega_{l, i} \Pi_{I_l, i}(x_i) + \left(1 - \sum_{l \in [k-1]} \omega_{l, i} \right) \Pi_{I_k, i}(x_i) \right].
\end{aligned}$$

The posterior $\Pi(I|X)$,

$$\begin{aligned}
\Pi(I|X) &= \frac{\lambda_I P_{X, I}}{P_X} = \frac{C_\lambda \prod_{l \in [k-1]} \prod_{i \in I_l} \lambda_{l, i} P_{X, I}}{P_X} \\
&= \frac{C_\lambda \bigotimes_{l \in [k-1]} \bigotimes_{i \in I_l} \lambda_{l, i} P_{X, I_l, i}(x_i) \times \bigotimes_{j \in I_k} P_{X, I_k, j}(x_j)}{P_X}.
\end{aligned}$$

The posterior $\Pi(\theta|X)$,

$$\begin{aligned}
\Pi(\theta|X) &= \sum_{I \in \mathcal{I}} \Pi_I(\theta|X) \Pi(I|X) = \sum_{I \in \mathcal{I}_k} \lambda_I \frac{\bigotimes_{l \in [k]} \bigotimes_{i \in I_l} \Pi_{I_l, i}(\theta_i) \cdot P_{\theta_i}(X_i)}{P_X} \\
&= \frac{1}{P_X} \sum_{I \in \mathcal{I}_k} C_\lambda \left[\bigotimes_{l \in [k-1]} \bigotimes_{i \in I_l} \lambda_{l, i} \Pi_{I_l, i}(\theta_i) \cdot P_{\theta_i}(X_i) \times \bigotimes_{j \in I_k} \Pi_{I_k, j}(\theta_j) \cdot P_{\theta_j}(X_j) \right] \\
\Pi(\theta|X) &= \frac{1}{P_X} \bigotimes_{i \in [n]} \sum_{l \in [k-1]} \omega_{l, i} \Pi_{I_l, i}(\theta_i) \cdot P_{\theta_i}(X_i) + \left(1 - \sum_{l \in [k-1]} \omega_{l, i} \right) \Pi_{I_k, i}(\theta_i) \cdot P_{\theta_i}(X_i).
\end{aligned}$$

□

General k-groups: cardinality-based

Proposition A.4 (Construction of cardinality-based prior). Assume prior λ on the partition space $\mathcal{I}_k$, $I \sim \lambda_I$ of the form $\lambda_I = c_{\mathbf{m}} g(\mathbf{m})$ where $g : \mathcal{M} \rightarrow [0, 1]$, is a nonnegative function s.t. $\sum_{\mathbf{m} \in \mathcal{M}} g(\mathbf{m}) = 1$ with,

$$\mathcal{M} = \{\mathbf{m} = (m_1, \dots, m_k) \in \mathbb{N}_0^k : m_l = |I_l|, \forall l \in [k], I = (I_1, \dots, I_k) \in \mathcal{I}_k\}.$$

Then the constants $c_{\mathbf{m}}$ must satisfy,

$$c_{\mathbf{m}} = \binom{n}{m_1, \dots, m_k}^{-1}, \quad \forall \mathbf{m} \in \mathcal{M}.$$

Proof. Suppose λ_I is a prior on $\mathcal{I}_k$ of the form $\lambda_I = c_{\mathbf{m}} g(\mathbf{m})$. Abusing notation $\mathbf{m} = |I|$ to be the vector $\mathbf{m} = (m_1 = |I_1|, \dots, m_k = |I_k|)$. Given that the number of partitions with $|I| = \mathbf{m}$ is equal to the multinomial coefficient $\binom{n}{m_1, \dots, m_k}$, it follows that,

$$\sum_{I \in \mathcal{I}_k} \lambda_I = \sum_{\mathbf{m} \in \mathcal{M}} \sum_{I: |I| = \mathbf{m}} c_{\mathbf{m}} g(\mathbf{m}) = \sum_{\mathbf{m} \in \mathcal{M}} \binom{n}{m_1, m_2, \dots, m_k} c_{\mathbf{m}} g(\mathbf{m}). \quad (\text{A.4})$$

By requiring $\sum_{I \in \mathcal{I}_k} \lambda_I = 1$ and since, $\sum_{\mathbf{m} \in \mathcal{M}} g(\mathbf{m}) = 1$, it must hold that,

$$c_{\mathbf{m}} = \binom{n}{m_1, m_2, \dots, m_k}^{-1}, \quad \forall \mathbf{m} = (m_1, \dots, m_k) \in \mathcal{M},$$

□

Proof of Lemma 2.12, Multivariate Polynomial Trick . Suppose $\psi_1, \dots, \psi_k \in C^2(\mathbb{R})$ evaluated at points $x_1, \dots, x_n \in \mathbb{R}$. Define the matrix $\mathbf{A} \in \mathbb{R}^{k \times n}$ by

$$a_{l,i} := \psi_l(x_i), \quad \text{for all } l \in [k], i \in [n]$$

Let $\mathbf{a}_l = (a_{l,1}, \dots, a_{l,n}) \in \mathbb{R}^n$ denote the l -th row of $\mathbf{A}$. Then, the degree- k homogeneous polynomial in $\mathbb{R}[Z_1, \dots, Z_n]$,

$$P_{\mathbf{A}}(Z) = \Phi(a_1, \dots, a_k) = \prod_{i=1}^n (a_{1,i} Z_1 + a_{2,i} Z_2 + \dots + a_{k,i} Z_k)$$

Its $\mathbf{m}^{th}$ -order coefficient, $A_{\mathbf{m}}$ is the coefficient of the monomial, $Z_1^{m_1} Z_2^{m_2} \dots Z_k^{m_k}$. Using the expansion of the polynomial, choose exactly m_1 indices, $i \in S_1 \subseteq [n]$, then choose m_2 indices, $i \in S_2 \subseteq [n - m_1]$, repeat till m_{k-1} , $i \in S_{k-1} \subseteq [n - (m_1 + \dots + m_{k-2})]$. It follows that $|S| = \mathbf{m}$ and

$$\prod_{i \in S_1} (a_{1,i} \cdot Z_1) \prod_{i \in S_2} (a_{2,i} \cdot Z_2) \dots \prod_{i \in S_k} (a_{k,i} \cdot Z_k) = \left(\prod_{i \in S_1} a_{1,i} \right) \dots \left(\prod_{i \in S_2} a_{2,i} \right) \left(\prod_{i \in S_k} a_{k,i} \right) Z_1^{m_1} \dots Z_k^{m_k}$$

where, $Z_1 = 1$.

Summing over all subsets $S_1, \dots, S_k \subseteq [n]$, s.t. $|S| = \mathbf{m}$, it holds that $S = (S_1, \dots, S_k)$ is a partition of $[n]$ indices into k -groups, $S \in \mathcal{I}_k$,

$$A_{\mathbf{m}}(a_1, \dots, a_k) = \sum_{|S|=\mathbf{m}} \left(\prod_{i \in S_1} a_{1,i} \prod_{i \in S_2} a_{2,i} \cdots \prod_{i \in S_k} a_{k,i} \right).$$

□

Prediction

Proof of Lemma 2.14. Given the posterior from the bayesian analysis from Theorem (2.9) and the elementary formula (A.3), it follows that,

$$\begin{aligned} p_{l^*,i} &= \sum_{I:i \in I_l} \Pi(I|X) = \sum_{I:i \in I_{l^*}} \frac{\bigotimes_{l \in [k]} \bigotimes_{j \in I_l} \omega_{l,j} P_{X,I_l,j}(x_j)}{P_X} \\ &= \frac{\omega_{l^*,i} P_{X,I_{l^*},i}(x_i)}{P_X} \sum_{I:i \in I_l} \bigotimes_{l \in [k]} \bigotimes_{j \in I_l, j \neq i} \omega_{l,j} P_{X,I_l,j}(x_j) \\ &\stackrel{(A.3)}{=} \frac{\omega_{l^*,i} \cdot P_{X,I_{l^*},i}(x_i)}{P_X} \bigotimes_{j \in [n], j \neq i} \left(\sum_{l \in [k]} \omega_{l,j} P_{X,I_l,j}(x_j) \right) \end{aligned}$$

By substituting $P_X = \prod_{i=1}^n \sum_{l=1}^k \omega_{l,i} P_{X,I_l,i}(x_i)$, we arrive at the final expression,

$$p_{l^*,i} = \frac{\omega_{l^*,i} \cdot P_{X,I_{l^*},i}(x_i)}{\sum_{l \in [k-1]} \omega_{l,i} P_{X,I_l,i}(x_i)} \quad \text{for } l^* \in [k].$$

□

Proof of Lemma 2.15. Given the posterior from the bayesian analysis from Theorem

(2.11) and the elementary formula (A.3), it follows that,

$$\begin{aligned}
p_{l^*,i} &= \sum_{I: i \in I_l} \Pi(I|X) = \sum_{I: i \in I_{l^*}} \lambda_I \frac{\bigotimes_{l \in [k]} \bigotimes_{i \in I_l} P_{X, I_l, i}(x_i)}{P_X} \\
&= \frac{1}{P_X} \sum_{m \in \mathcal{M}} \sum_{\substack{I: |I|=m \\ i \in I_{l^*}}} g(\mathbf{m}) \bigotimes_{l \in [k]} \bigotimes_{j \in I_l} P_{X, I_l, j}(x_j) \\
&= \frac{P_{X, I_{l^*}, i}(x_i)}{P_X} \sum_{m \in \mathcal{M}} \sum_{I: |I|=m} g(\mathbf{m}) \bigotimes_{l \in [k]} \bigotimes_{\substack{j \in I_l \\ j \neq i}} P_{X, I_l, j}(x_j)
\end{aligned}$$

Finally,

$$p_{l^*,i} = \frac{P_{X, I_{l^*}, i}(x_i)}{P_X} \sum_{m \in \mathcal{M}} g(\mathbf{m}) A_{\mathbf{m}_{l^*}, -i}(P_{X, I_1}(x), P_{X, I_2}(x), \dots, P_{X, I_k}(x)), \quad \text{for } l^* \in [k].$$

with,

$$\mathbf{m}_{l^*} = \begin{cases} (m_1, \dots, m_{l^*} - 1, \dots, m_k), & \text{for } l^* \in [k-1] \\ \mathbf{m} & l^* = k \end{cases}$$

□

A.2 Additional figures and tables

Cluster	Feature	SW Statistic	p-value	Normal
0	Sepal Area	0.9607	0.9433	Yes
1	Sepal Area	0.7247	0.9666	Yes
2	Sepal Area	0.9660	0.8511	Yes
3	Sepal Area	0.9713	0.9526	Yes
0	Petal Area	0.4282	0.1767	Yes
1	Petal Area	1.5960	0.5612	Yes
2	Petal Area	0.1675	0.00002	No
3	Petal Area	0.2495	0.0405	Yes

Table A.1: Shapiro–Wilk test results for normality of sepal and petal area distributions across clusters. A significance level of $\alpha = 0.00625$ is used.

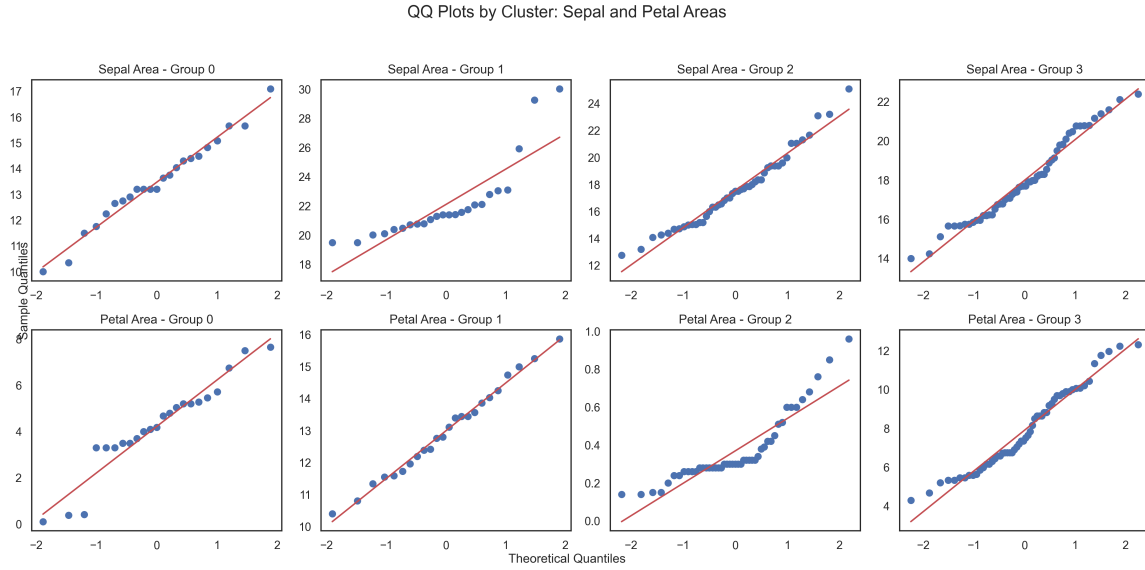


Figure A.1: QQ-plots for the assumptions check on the iris dataset, refer to section 4.2.1.

Cluster	Chi-squared	p-value	Significant
0	554.1998	0.0000	Yes
1	592.2089	0.0000	Yes
2	3604.5515	0.0000	Yes
3	1645.4726	0.0000	Yes

Table A.2: Chi-Squared Test for Variance by Cluster

A.3 Python code

The source code used to develop, test, and evaluate the Bayesian clustering methods presented in this thesis is publicly available at: [bayesiancl-repo]. The repository includes:

- The full implementation of the proposed Bayesian clustering algorithm.
- Scripts for synthetic data generation and benchmarking experiments.
- Evaluation metrics and comparison baselines (e.g., KMeans, DBSCAN, ProdForm).
- Instructions for reproducing the figures and results reported in Chapters 3 & 4.

The codebase is written in Python and includes configuration files to ensure full reproducibility of the experimental results. Installation and usage instructions can

be found in the accompanying README file.